Complex Numbers

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Chapter-7 Introduction to Imaginary and Complex Numbers

7.1 Imaginary Numbers

The square root of a negative number is undefined. Try taking the square root of -36 on most calculators and you will get an error message.

Now we are going to introduce a new type of number, the *imaginary* number. On the real number line the unit is "1". Just like real numbers, there is a unit of imaginary numbers. That unit is "*i*" (sometimes "*j*" is used as well, most often in engineering disciplines, but we will stick with "*i*"), and it has the property that

$$i^2 = -1. \qquad (1)$$

We can have more than one unit of imaginary numbers, or fractions of a unit of imaginary numbers. All of the following are valid imaginary numbers:

a) i b) 2i c) -4i d) 0.01i e) -0.025i f) πi

g) $2\sqrt{2}\ i$

and they all share the same property that they are some *multiple* of the unit imaginary number, *i*. Because of the property that *i* times itself is -1, the following rules follow:

- An imaginary number times a real number is an imaginary number.
- An imaginary number times an imaginary number is a real number.

We can show that this last statement is true by factoring out *i* when performing multiplication,

$$3i \quad 2i \;=\; 3\cdot 2 \quad i-i \;=6_i^2=-6\,. \qquad (2)$$

Similarly, we can write down the powers of *i*,

$$\begin{aligned} i^1 &= i \\ i^2 &= -1 \\ i^3 &= -i \\ i^4 &= +1 \end{aligned} \qquad (3)$$

and so forth.

7.2 Complex Numbers and Complex Arithmetic

More interesting (and more useful) than imaginary numbers are *complex* numbers—numbers that are a combination of real and imaginary elements. These are referred to as the *real part* and the *imaginary part* of the complex number. Complex numbers are formed by adding real and imaginary numbers together, such as $3+2i$.

More than two numbers added together are also a complex number, as long as there is at least one each of imaginary and real types. All of the real elements together are the real part of the complex number, and all of the imaginary elements together are the imaginary part. Usually we collect these parts together to make it clear which element belongs to which part,

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$$1+2i-3+4i+5=(1-3+5)+(2+4)i$$

real part — imaginary part

$$=3+6i \qquad (4)$$

The real and imaginary parts of a complex number are separate (they are on separate number lines), and this leads us to the first important rule of complex arithmetic:

- When adding or subtracting complex numbers, add or subtract the real and imaginary parts separately.

An example of complex addition is

$$\begin{array}{r} 3+4i \\ 2+3i \\ \hline 5+7i \end{array} \qquad (5)$$

where, again, the real and imaginary components are treated separately.

Multiplication, on the other hand, involves both parts of the complex number. The rule is:

- When multiplying a complex number by a real (or imaginary) number, multiply both the real and the imaginary parts of the complex number by the real (or imaginary) number.

A similar rule applies to multiplying two complex numbers:

- When multiplying two complex numbers, multiply both the real and imaginary parts of both numbers by the real and imaginary parts of the other.

This rule is usually easiest to apply using the FOIL (First, Outer, Inner, Last) method. A couple of examples are

$$3\ 5+4i\ =3.5+34i=15+20i \qquad (6)$$

and

$$\begin{aligned} 3-2i\ \ 5+4i\ \ &=3\cdot 5+3\cdot 4i+\ -2i\ 5+\ -2i\ \ 4i \\ &=15+12i-10i+8 \\ &=23+2i \end{aligned} \qquad (7)$$

Notice in the last step the real and imaginary parts of the complex result have been collected together.

7.2.1 Complex Numbers as Vectors

Complex number do not lie on the real number line, nor on the imaginary number line, but they have one foot in each camp. It is traditional to represent complex numbers as 2-dimensional vectors, where the real part is on the horizontal axis and the imaginary part on the vertical axis. Figure 1 shows this imaginary plane with several numbers (pure real, pure imaginary, and complex) represented as vectors. Pure real numbers are vectors in the horizontal direction, either positive or negative. Similarly, pure imaginary numbers are vectors in the vertical direction, either positive or negative.

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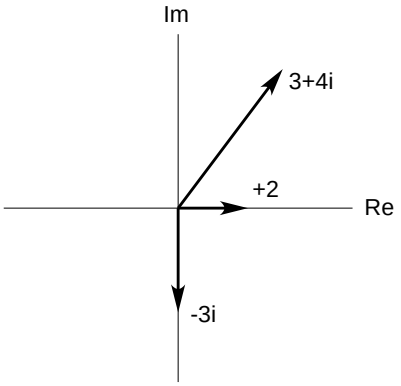


Figure: The complex plane with several numbers represented as vectors.

Like any other vectors, complex numbers have a magnitude and direction. The magnitude of the complex number *A* is not the magnitude of its real and imaginary parts added together. Rather, it is

$$|A| = \sqrt{|\text{Re } A|^2 + |\text{Im } A|^2} \tag{8}$$

where Re(*A*) and Im(*A*) are the real and imaginary parts of *A*, respectively. For example,

$$|3+4i| = \sqrt{3^2+4^2} = \sqrt{9+16} = \sqrt{25} = 5 \tag{9}$$

This is really nothing other than the Pythagorean theorem, applied to the complex plane, as Figure 2 illustrates.

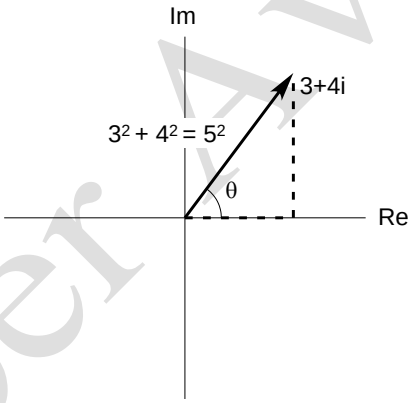


Figure: Like any other vector, complex numbers on the complex plane have magnitude and direction.

The direction of a vector (complex number) on the complex plane is defined as the angle θ between the vector and the real (horizontal) axis, as shown in Figure 3. This angle can be determined from the equation

$$. \theta = \tan^{-1}\left(\frac{\text{Im } A}{\text{Re } A}\right) \tag{10}$$

7.2.2 The Complex Exponential

There is one set of notations associated with complex numbers that is potentially confusing because it looks very similar to notation used with real numbers, while meaning something entirely different. This is the complex exponential. The transcendental number e raised to a (real) power is just another real number, for example

$$e^2 = 7.389056.... \tag{11}$$

If we raise *e* to an imaginary power, however, we have something different entirely. By definition, *e* raised to the imaginary power θi is

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$$. \; e^{\theta i} = \cos\ \theta\ + i \sin\ \theta \qquad (12)$$

This is a surprisingly useful definition. If $\theta = 0$, this is simply the real number 1. If $\theta = \pi$, this is the pure imaginary number *i*. Whatever the value of θ, the magnitude of $e^{\theta i}$ is always 1, as we can see from using the definition from Equation (12) in

Equation (8),

$$\left|e^{\theta i}\right| = \sqrt{\cos^2\ \theta\ + \sin^2\ \theta\ } = \sqrt{1} = 1 \left|e\theta\right| \qquad (13)$$

As a matter of fact, $e^{\theta i}$ represents a unit vector that spins around the axis as the angle θ increases. If, instead of a plain angle, we take the angle to be $\theta = \omega t$, we have a vector spinning with an angular frequency ω. Multiplying this unit vector by a constant *A* and taking the real part, we have

$$\begin{aligned} \text{Re}\ Ae^{i\omega t} \quad &= \text{Re}\left[A\cos\ \omega t\ + iA\sin\ \omega t\ \right] \\ &= A\cos\ \omega t \end{aligned} \qquad (14)$$

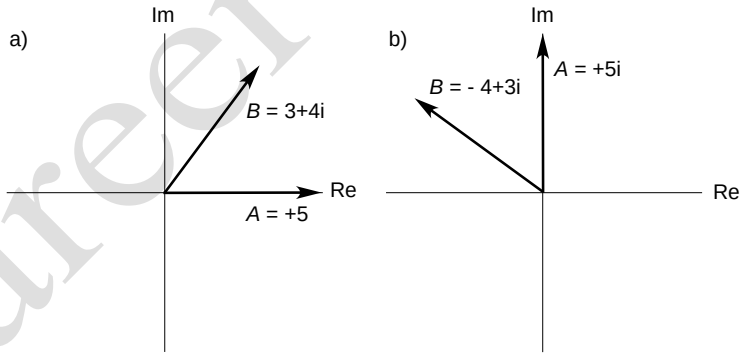
which is a sinusoidal waveform. So, is the complex exponential just a mathematical shorthand? Yes and no. It is another way of expressing an oscillating waveform. The great advantage of this notation is that the amplitude *A* in Equation (14) is not restricted to being a real number. Consider two complex exponentials, one with a real amplitude, one with a complex amplitude:

$$A = 5e^{i\omega t} \qquad B = \ 3 + 4i\ e^{i\omega t} \qquad (15)$$

If $t = 0$, then $e^{i\omega t} = \cos(0) + i\sin(0) = 1$, and

$$A = 5 \qquad B = 3 + 4i \qquad (16)$$

These are just the same examples we used before, shown again here in Figure 3(a). Both have a magnitude (length) of 5, but they are oriented differently in the complex plane.



(a) Two vectors, both with the same magnitude, one with a real amplitude, the other with a complex amplitude, at time $t = 0$.

(b) The same two vectors at time t *picked so that* $\omega t = \pi/2$.

The two maintain a constant phase difference as they spin around in the complex plane.

Now, if we pick the time *t* so that $\omega t = \pi / 2$, then $e^{i\omega t} = \cos\ \pi / 2\ + i \sin\ \pi / 2\ = i$, and

$$\begin{aligned} A &= 5i \qquad & B &= \ 3 + 4i\ i \\ & & &= -4 + 3i \end{aligned} \qquad (17)$$

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If we look at these plotted in the complex plane (Figure 3(b)), then we see that the two vectors have maintained the same orientation relative to each other, but that both have turned 90°. The constant difference between the two is like a phase angle ϕ. If we use a complex amplitude *A*, then we have two ways of writing a sinusoidal waveform,

$$\text{Re}\ Ae^{i\omega t} = |A|\cos\ \omega t + \phi \qquad (18)$$

The difference between the left-hand-side and the right-hand-side of Equation (18) is that the left-hand-side has one fewer variable than the right-hand-side. Remember that it takes *N* equations to solve for *N* unknown variables in a system of equations. This mathematical shorthand has reduced the number of variables in the problem. For some problems, this makes all the difference in the world in whether the problem can be solved at all or not!

7.2.3 Complex Numbers: Fundamental Concepts

(1) A ***complex number*** z is a number of the form a + bi where a, b are real numbers and $i^2 = -1$.

(2) The set **C** of all complex numbers is defined by **C**= $a + bi : a.b \in R \text{ and } i^2 = -1$

Where a is called the ***real part*** of z and a = Re (z) and

b is called the ***imaginary part*** of z and b = Im (Z).

(3) z is said to be ***purely imaginary*** if and only if Re (z) = 0 and Im (z) $\neq 0$.

(4) When Im (z) = 0, the complex number z is ***real***.

N.B. $i^3 = i^2 \cdot i = -1, \quad i^4 = i^2 \cdot i^2 = 1, \quad i^5 = i^4 \cdot i = i, \quad i^6 = i^4 \cdot i^2 = -1$.

Operations On Complex Numbers

Let z_1 = a+ bi and z_2 = c + di. Then

(1) $z_1 + z_2 = \ a + c \ + \ b + d \ i$

(2) $z_1 - z_2 = \ a - c \ + \ b - d \ i$

(3) $z_1 z_2 = \ a + bi \quad c + di$

$= \ ac - bd \ + \ ad + bc \ i$

(4) $\frac{z_1}{z_2} = \frac{a+bi}{c+di} \cdot \frac{c-di}{c-di} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2} i$, where $z_2 \neq 0$.

N.B. (i) $\frac{1}{i} = \frac{i}{i^2} = \frac{i}{-1} = i$;

(ii) $\frac{1}{z_2} = \frac{1}{c+di} = \frac{c-di}{c^2+d^2} = \frac{1}{c^2+d^2}\overline{z_2}$.

Some definitions

Suppose that *z* is a complex number. Let *z* = *a* + *ib*.

The **real part** of *z*, written Re(*z*), is *a*.

The **imaginary part** of *z*, written Im(*z*), is *b*.

The **complex conjugate** of *z*, written *z** or $\overline{z}$, is *z** = *a* – *ib*.

Example 1: Let *z* = 5 – 2i, *w* = -2 + i and *u* = 7i.

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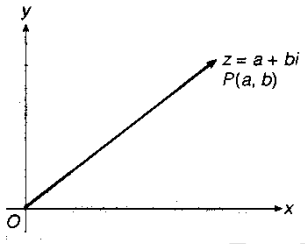
Then:

Re(z) = 5	Im(z) = -2	z^* = 5 + 2i
Re(w) = -2	Im(w) = 1	w* = -2 – i.
Re(u) = 0	Im(u) = 7	u^* = -7i.

7.2.4 Geometrical Representation of a Complex Number

From the definition of complex numbers, a complex number $z = a + bi$ is defined by the two real numbers a and b. Hence, if we consider the real part a as the x - coordinate in the rectangular coordinates system and the imaginary part b as the y - coordinate then the complex number z can be represented by the point (a,b) on the plane. This plane is called the ***complex plane*** or the ***Argand diagram***.

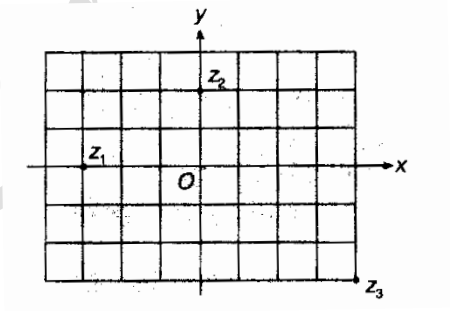
On this plane, real numbers are represented by points on x – axis which is called the ***real axis***; imaginary numbers are represented by points on the y- axis which is called the ***imaginary axis***. The number 0 is represented by the origin ***O***.



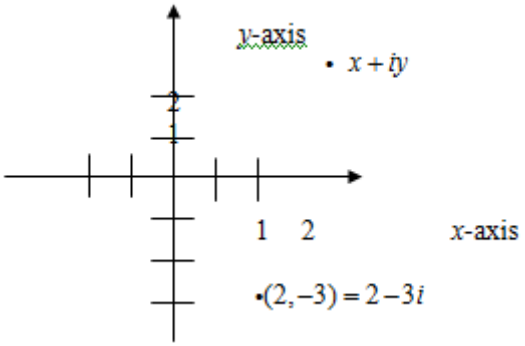
Any point (a,b) on this plane can be used to represent a complex number z = a + bi.

For example, as shown in Figure, the z_1, z_2, z_3 represents respectively the complex numbers.

$z_1 = -3, z_2 = 2i, z_3 = 4 - 3i$



The geometric representation of complex numbers is to represent the complex number (x, y) as the point (x, y)



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So the real number (x., 0) is the point on the horizontal *x*-axis, the purely imaginary number yi = (0,y) is on the vertical y-axis. For the complex number (x,y), x is the real part and *y* is the imaginary part.

How do we **divide** complex numbers? Let's introduce the conjugate of a complex number then go to division.

Given the complex number z = x + iy, define the conjugate $\bar{z} = \overline{x+iy} = x - iy$

We can divide by using the following:

$$\frac{z_1}{z_2} = \frac{x_1+iy_1}{x_2+iy_2} = \frac{x_1+iy_1}{x_2+iy_2}\frac{x_2-iy_2}{x_2-iy_2} = \frac{x_y x_2 + y_1 y_2 + i\ x_2 y_1 - x_1 y_2}{x_2^2+y_2^2}\frac{z_1}{z_2}$$

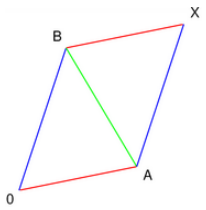
Example 5. $\frac{2+3i}{3-4i} = \frac{(2+3i)(3+4i)}{(3-4i)(3+4i)} = \frac{6+12i^2+8i+9i}{9-16i^2} = -\frac{6}{25} + i\frac{17}{25}$

7.2.4.1 Geometrical Applications

	Vectors	Complex Numbers
Addition	$a\vec{i}+b\vec{j} + c\vec{i}+d\vec{j}$ $= a+c\ \vec{i} + b+d\ \vec{j}$	$a+bi + c+di$ $= a+c + b+d\ i$
Subtraction	$a\vec{i}+b\vec{j} - c\vec{i}+d\vec{j}$ $= a-c\ \vec{i} + b-d\ \vec{j}$	$a+bi - c+di$ $= a-c + b-d\ i$
Scalar multiplication	$\lambda\ a\vec{i}+b\vec{j} = \lambda a\vec{i}+\lambda b\vec{j}, \forall \lambda \in R$	$\lambda\ a+bi = \lambda a+\lambda bi, \forall \lambda \in R$

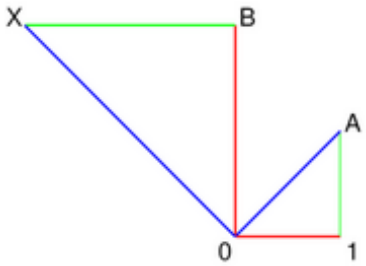
In the set **C** of all complex numbers, if z = a + bi is regarded as a vector v = ai + bj ; then as far as the above three operations are concerned complex numbers behave similar to those of vectors.

Geometric interpretation of the operations on complex numbers



The point X is the sum of A and B.

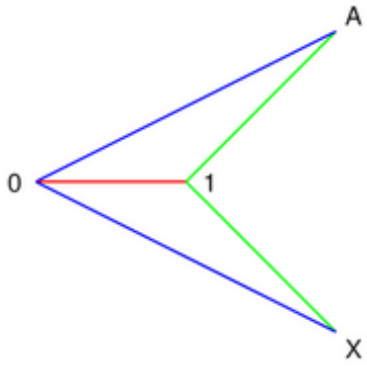
Choose a point in the plane which will be the origin, 0. Given two points A and B in the plane, their sum is the point X in the plane such that the triangles with vertices 0, A, B and X, B, A are similar.



The point X is the product of A and B.

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Choose in addition a point in the plane different from zero, which will be the unity, 1. Given two points A and B in the plane, their product is the point X in the plane such that the triangles with vertices 0, 1, A, and 0, B, X are similar.



The point X is the complex conjugate of A.

Given a point A in the plane, its complex conjugate is a point X in the plane such that the triangles with vertices 0, 1, A and 0, 1, X are mirror image of each other.

7.2.4.2 Complex Numbers in Polar Form

It is possible to express complex numbers in polar form. If the point z = (x,y) = x + iy is represented by polar coordinates r,θ, then we can write $x = r\cos\theta$, $y = r\sin\theta$ and $z = r\cos\theta + ir\sin\theta = re^{i\theta}$. r is the modulus or absolute value of $z, |Z| = r = \sqrt{x^2 + y^2}$ and θ is z, the argument of z, $\theta = \arctan\left(\frac{y}{x}\right)$. The values of *r* and θ determine *z* uniquely, but the converse is not true. The modulus *r* is determined uniquely by *z*, but θ is only determined up to a multiple of 2π. There are infinitely many values of θ which satisfy the equations $x = r\cos\theta, y = r\sin\theta$, but any two of them differ by some multiple of 2π. Each of these angles θ is called an **argument** of *z*, but, by convention, one of them is called the **principal argument**.

Definition: If *z* is a non-zero complex number, then the unique real number θ, which satisfies

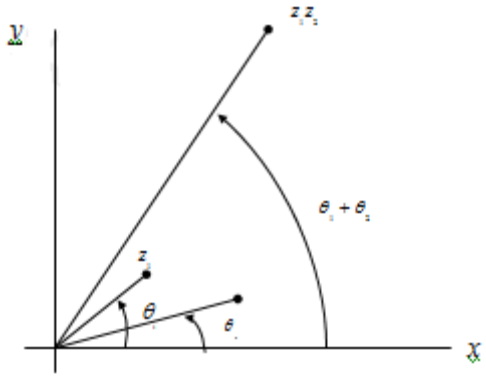
$$x = |z|\cos\theta, y = |z|\sin\theta, -\pi < \theta \le \pi$$

is called the **principal argument** of *z*, denoted by $\theta = \arg\ z$.

Note: The distance from the origin to the point (x,y) is |z|, the modulus of *z*; the argument of *z* is the angle $\theta = \arctan\frac{y}{x}$ Geometrically, θ is the directed angle measured from the positive *x*-axis to the line segment from the origin to the point (x,y). When z = 0, the angle θ is undefined.

The polar form of a complex number allows one to multiply and divide complex number more easily than in the Cartesian form. For instance, if $z_1 = r_1 e^{i\theta_1}$ and $z_2 = r_2 e^{i\theta_2}$ then $z_1 z_2 = r_1 r_2 e^{i\,\theta_1+\theta_2}$, $\frac{z_1}{z_2} = \frac{r_1}{r_2} e^{i\,\theta_1-\theta_2}$ .These formulae follow directly from DeMoivre's formula.

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Example 6. For z = 1 + i, we get $r=\sqrt{1^2+1^2}=\sqrt{2}$ and $\theta=\arctan\frac{y}{x}=\arctan 1=\frac{\pi}{4}$. The principal value of θ is $\frac{\pi}{4}$, but $\frac{9\pi}{4}$ would work also.

Multiplication and Division in Polar Form

Let $z_1=r_1\cos\theta_1+ir_1\sin\theta_1=r_1\ \cos\theta_1+i\sin\theta_1$ and $z_2=r_2\ \cos\theta_2+i\sin\theta_2$ then we have

$z_1z_2=r_1r_2\ \cos\ \theta_1+\theta_2\ +i\sin\ \theta_1+\theta_2$ and $\frac{z_1}{z_2}=\frac{r_1}{r_2}\cos\ \theta_1-\theta_2\ +\sin\ \theta_1-\theta_2$

Example 7: $z_1=1+i=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$ and $z_2=\sqrt{3}-i=2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$

Then $z_1z_2=\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)=2\sqrt{2}\left(\cos\frac{5\pi}{12}+i\sin\frac{5\pi}{12}\right)$

Since $\frac{\pi}{4}+\frac{\pi}{6}=\frac{10\pi}{24}=\frac{5\pi}{12}$

And

$$\frac{z_1}{z_2}=\frac{\sqrt{2}\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)}{2\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)}=\frac{\sqrt{2}}{2}\left(\cos\frac{\pi}{12}+i\sin\frac{\pi}{12}\right)$$

We can use $z^2=z\cdot z=r\cdot r\ \cos\ \theta+\theta\ +i\sin\ \theta+\theta\ =r^2\ \cos 2\theta+i\sin 2\theta$

7.2.5 DeMoivre's Theorem:

$$z^n=r^n\ \cos n\theta+0\sin n\theta$$

where *n* is an positive integer.

Let r = 1 to get: $\cos\theta+\sin\theta\ "=\cos n\theta+i\sin n\theta$.

DeMoivre's Theorem and nth Roots of a Complex Number

For any real number $\theta_1,\theta_2,\ \cos\theta_1+i\sin\theta_1\ \ \cos\theta_2+i\sin\theta_2\ =\cos\ \theta_1+\theta_2\ +i\sin\ \theta_1+\theta_2$

In particular, if $\theta=\theta_1=\theta_2$, we have $\cos\theta+i\sin\theta\ ^2=\cos 2\theta+i\sin 2\theta$.

For any positive integer n, by induction on n, the result may be generalized as

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$$\cos\theta + i\sin\theta\ "= \cos n\theta + i\sin n\theta \quad \cdots\cdots\cdots(*)$$

and this is known as the DeMoivre's Theorem for integral index.

For any negative integer n , we may let n = - m. Then

$$\begin{aligned}
\cos\theta + i\sin\theta\ " &= \cos\theta + i\sin\theta\ ^{-m} \\
&= \frac{1}{\cos\theta + i\sin\theta\ ^{m}} \\
&= \frac{1}{\cos m\theta + i\sin m\theta} \\
&= \frac{\cos m\theta - i\sin m\theta}{\cos^2 m\theta + \sin^2 m\theta} \\
&= \cos\ -m\theta\ + i\sin\ -m\theta \\
&= \cos n\theta + i\sin n\theta
\end{aligned}$$

Hence, $(*)$ also holds for negative integers n .

For any rational number n. Put $n = \frac{p}{q}$, where $p, q \in Z$ and no loss of generality if q is taken as

to be positive. Then

$$\begin{aligned}
\cos n\theta + i\sin n\theta\ ^{q} &= \cos nq\theta + i\sin nq\theta \\
&= \cos\ p\theta + i\sin\ p\theta \\
&= \cos\theta + i\sin\theta\ ^{P} \\
\therefore \cos n\theta + i\sin n\theta\ \therefore &= \cos\theta + i\sin\theta\ ^{\frac{p}{q}} \\
&= \cos\theta + i\sin\theta\ ^{n}
\end{aligned}$$

In general, for any real number θ , positive number r and rational number n, we have

$$\left[r\ \cos\theta + i\sin\theta\ \right]^n = r^n\ \cos n\theta + i\sin n\theta$$

The nth roots of a complex number w are the n values of z which satisfy the equation $z^n = w$.

If we write $w = \cos\theta + i\sin\theta$ and assuming that the equation is satisfied by, $z = \cos\phi + i\sin\phi$

then

$$\cos\theta + i\sin\theta\ ^{\frac{1}{n}} = \cos\phi + i\sin\phi \quad \cdots\cdots\cdots(*)$$

$$\cos\theta + i\sin\theta = \cos\phi + i\sin\phi\ ^{n} = \cos n\phi + i\sin n\phi$$

By equating the real parts and imaginary parts on both sides, we have

$$\cos\theta = \cos n\phi \qquad \sin\theta = \sin n\phi$$

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$$n\phi = 2k\pi + \theta \qquad \phi = \frac{2k\pi + \theta}{n} \qquad \cdots\cdots\cdots (**)$$

where $k \in Z$. .

For $k = 0,1,2,...,n-1$, since $\frac{\theta}{n} \le \frac{2k\pi + \theta}{n} \le \frac{2n-2}{n}\pi + \frac{\theta}{n} < 2\pi + \frac{\theta}{n}$

We obtain the n distinct complex roots for $(*)$ with the values of ϕ obtained in $(**)$.

For $k < 0$ or $k > n-1$, the root obtained is equal to one of the roots mentioned above. Hence, the equation

$(*)$ has only n distinct complex roots.

7.2.5.1 DeMoivre's Theorem for Rational Index

Let n be a positive integer and θ be a real number. Then

$\cos\theta + i\sin\theta\ ^{\frac{1}{n}} = \cos\frac{2k\pi + \theta}{n} + i\sin\frac{2k\pi + \theta}{n}$, where $k = 0,1,2,...,n-1$

Using the DeMoivre's Theorem, we will have the following properties.

(i) $z^m\ z^n = z^{m+n}$

(ii) $\frac{z^m}{z^n} = z^{m-n}$

(iii) $z^0 = 1$

(iv) $z^{m\ n} = z^{mn}$

7.2.5.2 Application of DeMoivre's Theorem in Trigonometry

Direct application of DeMoivre's Theorem and the binomial theorem, we are able to express

(i) multiple angles such as $\sin n\theta$ and $\cos n\theta$ in terms of $\sin\theta$ and $\cos\theta$, and

(ii) powers of $\sin\theta$ and $\cos\theta$ back again into multiple angles.

Example 8: Compute $1+i^{\ 6}$

$$\begin{aligned} 1+i^{\ 6} &= \left(\sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)\right)^6 \\ &= \sqrt{2}^6\left(\cos 6.\frac{\pi}{4} + i\sin 6.\frac{\pi}{4}\right) \\ &= 8\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right) \\ &= -8i \end{aligned}$$

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7.2.6 Roots of Complex Numbers:

Consider $z = r\ \cos\theta + i\sin\theta\ = w^n = R^n\ \cos n\phi + i\sin n\phi$ (Equation 1)

where $w = R\ \cos\phi + i\sin\phi$. Then $R = \sqrt[n]{r}$, and so $\theta = n\phi$ or $\phi = \frac{\theta}{n}$.

However $n\phi = \theta + 2\pi$ also satisfies Equation 1 and so $\phi = \frac{\theta}{n} + \frac{2\pi}{n}$. And

$n\phi = \phi + 4\pi$ implies $\phi = \frac{\theta}{n} + \frac{4\pi}{n}$. However $n\phi = \theta + 6\pi$ implies $\phi = \frac{\theta}{n} + \frac{6\pi}{n}$.

And continuing $n\phi = \theta + k\pi$ implies $\phi = \frac{\theta}{n} + \frac{k\pi}{n}$. for k any integer up to n.

We get $\sqrt[n]{z} = \sqrt[n]{x}\left\{\cos\left(\frac{\theta + k2\pi}{n}\right) + i\sin\left(\frac{\theta + k2\pi}{n}\right)\right\}$, k=0, 1, 2, 3, •••, (n-1).

The nth roots of a Complex Number

If $w^n = z = r\ \cos\theta + i\sin\theta\ w^n$, then $w = r^{\frac{1}{n}}\left(\cos\frac{\theta + 2k\pi}{n} + i\sin\frac{\theta + 2k\pi}{n}\right)$, .k = 0,1,2,…n-1

nth of Unity

Theorem: nth of Unity of Their Properties

Let n be a positive integer. Then the equation $z^n = 1$ has n distinct roots given by

$$z_k = \cos\frac{2k\pi}{n} + i\sin\frac{2kn}{n} \quad k = 0,1,2...,,n-1$$

These roots are called the nth roots of unity. If we denote one of them by $\omega \neq 1$, then we have

$$\omega^n = 1$$

$$1 + \omega + \omega^2 + ... + \omega^{n-1} = 0$$

Example 9: Find the square roots of *i*. Since $i - \cos\left(\frac{\pi}{2}\right) + i\sin\left(\frac{\pi}{2}\right)$, we let

$w_1 = \sqrt{1}\left(\cos\left(\frac{1}{2}.\frac{\pi}{2}\right) + i\sin\left(\frac{1}{2}.\frac{\pi}{2}\right)\right) = \cos\frac{\pi}{4} + i\sin\frac{\pi}{4} = \frac{1}{\sqrt{2}} + i\frac{1}{\sqrt{2}}$ is one square root of i . The second square root of i is :

$w_1 = \sqrt{1}\left(\cos\left(\frac{\pi}{4} + \pi\right) + i\sin\left(\frac{\pi}{4} + \pi\right)\right) = \cos\frac{5\pi}{4} = \frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}$.

Example 10. Find the sixth root of -1.

There will be six roots:

$$z_1 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6}\right) + i\sin\left(\frac{\pi}{6}\right)\right) = \frac{\sqrt{3}}{2} + \frac{1}{2}i$$

$$z_2 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6} + \frac{2\pi}{6}\right) + i\sin\left(\frac{\pi}{6} + \frac{2\pi}{6}\right)\right) = i$$

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$$z_3 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6}+\frac{4\pi}{6}\right)+i\sin\left(\frac{\pi}{6}+\frac{4\pi}{6}\right)\right)=-\frac{\sqrt{3}}{2}+\frac{1}{2}i$$

$$z_4 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6}+\frac{6\pi}{6}\right)+i\sin\left(\frac{\pi}{6}+\frac{6\pi}{6}\right)\right)=-\frac{\sqrt{3}}{2}-\frac{1}{2}i$$

$$z_5 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6}+\frac{8\pi}{6}\right)+i\sin\left(\frac{\pi}{6}+\frac{8\pi}{6}\right)\right)=-i$$

$$z_6 = \sqrt[6]{1}\left(\cos\left(\frac{\pi}{6}+\frac{10\pi}{6}\right)+i\sin\left(\frac{\pi}{6}+\frac{10\pi}{6}\right)\right)=-\frac{\sqrt{3}}{2}-\frac{1}{2}i$$

Example 11: Compute $\sqrt[4]{-4i}$

Solution: $\sqrt[4]{-4i}=\sqrt[4]{4\left(\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}\right)}=\sqrt{2}\ \cos\phi+i\sin\phi$

Where $\phi=\frac{3\pi+k\cdot 2\pi}{2.4}.So\,\phi=\frac{3\pi}{8},\frac{7\pi}{8},\frac{11\pi}{8},\frac{15\pi}{8}$

$$\sqrt[4]{-4i}=\sqrt{2}\left(\cos\left(\frac{3\pi}{8}\right)+i\sin\left(\frac{3\pi}{8}\right)\right)=\sqrt{2}\ \ 0.3827+0.9239i$$

$$=\sqrt{2}\left(\cos\left(\frac{7\pi}{8}\right)+i\sin\left(\frac{7\pi}{8}\right)\right)=\sqrt{2}\ \ -0.9239+0.3827i$$

$$=\sqrt{2}\left(\cos\left(\frac{11\pi}{8}\right)+i\sin\left(\frac{11\pi}{8}\right)\right)=-\sqrt{2}\ \ 0.3827+0.9239i$$

$$=\sqrt{2}\left(\cos\left(\frac{15\pi}{8}\right)+i\sin\left(\frac{15\pi}{8}\right)\right)=\sqrt{2}\ \ -0.9239+0.3827i$$

7.3 Functions, Neighborhoods and Limits

We consider the concept of a function of a complex number. For y = f (x) = x^2 , where x and y are real numbers, we know about limits, continuity and derivatives. Let the complex number w = f (z) = z^2, where $z = x + iy$ and $w = \ x+iy\ ^2 = x^2+y^2+2xyi$. Here *x* and *y* are independent variables and w = u +iv where u = x^2+y^2 and v = 2xy. This is an example of a complex valued function, w, of a complex variable z. In general, let

w = u + iv where u = u (x,y) and v = v (x,y)



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Distance in the complex plane.

In the complex plane let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ then

$$|z_1 - z_2| = \sqrt{x_1 - x_2\ ^2 + y_1 - y_2\ ^2}$$

is the distance between the complex numbers z_1 and z_2 Note that $|z| = \sqrt{x^2 + y^2}$ is the distance from the origin to z.

Definition: A **neighborhood** $N\ z_0, r$ of the point z_0 in the complex plane is the set of points z where $r > 0$ and $z \| z - z_0 | < r$

Definition: A function w = f (z) is said to have a limit L as z approaches z_0 (and written $\lim_{z \to z_0} f\ z = L$) iff f (z) is defined in a neighborhood N (z_0,r), r > 0 except at z_0 and the values of f are close to L for all z close to z_0. Mathematically: for every $\varepsilon > 0$, we can find $\delta > 0$ such that for all $z \neq z_0$ in the neighborhood $N\ z_0, r$, so that if $0 < | z - z_0 | < \delta$, then $\left| f\ z - L \right| < \varepsilon$.

Notice that this definition of a limit is similar to the definition in calculus. The difference here is that *z* can approach z_0 from any direction in the complex plane.

Definition: A function $w = f\ z$ is **continuous** at z = z_0 if f (z_0) is defined and $\lim_{z \to z_0} f\ z = L$.

By definition a complex function that is continuous at z_0 is defined in a neighborhood of z_0. F (z) is continuous in a domain D if it is continuous at each point in the domain D.

The **derivative** of a function f (z) at a point *z* is defined as:

$$f'\ z = \lim_{h \to 0} \frac{f\ z + h - f\ z}{h}$$

provided that limit exists.

Example 12: Let $f(z) = z^2$. To compute $f'(z)$ we consider

$$\lim_{h \to 0} \frac{(z+h)^2 - z^2}{h} = \lim_{h \to 0} \frac{z^2 + 2zh + h^2 - z^2}{h} = \lim_{h \to 0} \frac{2zh + h^2}{h} = \lim_{h \to 0} (2z + h) = 2z$$

The rules for derivatives are the same for calculus:

$cf\ ' = cf\ '$ c=constant

$f + g\ ' = f\ ' + g\ '$

$fg\ ' = f\ 'g + fg\ '$

$$\left(\frac{f}{g} \right)' = \frac{f\ 'g - fg\ '}{g^2}$$

Definition: A function f(z) is defined to be **analytic in the domain D** if f (z)is defined and has a derivative at all points of D.

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The Exponential Function e^z

The definition of the exponential function $e^z = e^{x+iy} = e^x\, e^{iy}$ is given in terms of the real functions e^x, sin y, and cos y

$$e^z = e^{x+iy} = e^x\, e^{iy} = e^x\ \cos y + i \sin y$$

To show why this is true, consider the power series expansion: $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ... + \frac{x^n}{n!} + ...$

and substitute iy for *x*, to get:

$$e^{iy} = 1 + iy + \frac{iy}{2!}^2 + \frac{iy}{3!}^3 + ... + \frac{iy}{n!}^n + ... = 1 + iy - \frac{y^2}{2!} - \frac{iy^3}{3!} + ... + \frac{iy}{n!}^n + ...$$

$$e^{iy} = \left(1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \frac{y^6}{6!} + ...\right) + i\left(y - \frac{y^3}{3!} + \frac{y^5}{5!} - \frac{y^7}{7!} + ...\right) = \cos y + i \sin y\,.$$

The formula $e^z = e^{x+iy} = e^x\ \cos y + i \sin y$ is known as **Euler's formula**.

Theorem: e^z is never zero.

Proof: $e^z \cdot e^{-z} = e^0 = 1$**,** therefore, e^z cannot be zero.

Theorem: If *y* is real, then $\left|e^{iy}\right| = 1$.

Proof: $\left|e^{iy}\right|^2 = \cos^2 y + \sin^2 y = 1$ and $\left|e^{iy}\right| > 0$, which implies $\left|e^{iy}\right| = 1$.

Theorem: $e^z = 1$ if and only if $z = 2n\pi i$, where *n* is an integer.

Proof: If $z = 2n\pi i$, $e^z = \cos 2n\pi + i \sin 2n\pi = 1$ Conversely, suppose $e^z = 1$, then $e^x \cos y = 1, e^x \sin y = 0$. Since $e^x \neq 0$, this implies sin y = 0; and $y = k\pi$ where *k* is an integer. Substituting this into $e^x \cos y = 1$ implies $e^x\ -1\ ^k = 1$. Hence *k* is even, since $e^x > 0$. Therefore, $e^x = 1 and\ x = 0$.

Theorem: $e^{z_1} = e^{z_2}$ if and only if $z_1 - z_2 = 2n\pi i$, where *n* is an integer.

Proof: $e^{z_1} - e^{z_2}$ if and only if $e^{z_1 - z_2} = 1$. Then use the previous theorem.

Definition of sin z **and cos z**

To get the definitions of sin z and cos z we substitute -y for *y* in Euler's Formula to get:

$$e^{-iy} = \left(1 - \frac{y^2}{2!} + \frac{y^4}{4!} - \frac{y^6}{6!} + ...\right) + i\left(-y + \frac{y^3}{3!} - \frac{y^5}{5!} + \frac{y^7}{7!} - ...\right) = \cos y - i \sin y.$$

Adding e^{iy} and e^{-iy} together and dividing by 2, we get:

$$\frac{e^{iy} + e^{-iy}}{2} = \cos y$$

Subtracting e^{-iy} from e^{iy}, and dividing by 2i, we get: $\sin y = \frac{e^{iy} - e^{-iy}}{2i}$.

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We now define $\sin z = \frac{e^{iz} - e^{-iz}}{2i}$ and $\cos z = \frac{e^{iz} + e^{-iz}}{2}$.

Substitute *ix* for y in $\cos y = \frac{e^{iy} + e^{-iy}}{2}$ to get $\cos\ ix\ = \frac{e^{-x} + e^{x}}{2} = \cos h\, x$.

Similarly we get sin (ix) = I sin h x.

Facts:

1. For $z = x + 0i$, to get $e^{z} = e^{x-i0} = e^{x}$

2. The derivative of e^{z} is e^{z}.

3. $e^{z_1+z_2} = e^{z_1}\, e^{z_2}$

4. For z = iy we get $e^{iy} = \cos y + i \sin y$.

5. In polar form $z = x + iy = r\ \cos\theta + i \sin\theta\ = re^{i\theta}$.

6. It now follows that $e^{z+2\pi i} = e^{z}\, e^{2\pi i} = e^{z}.1 = e^{z}$

7. $e^{\pi i} = -1$

8. $\cos h\, z = \frac{e^{z} + e^{-z}}{2} \sin h\, z = \frac{e^{z} - e^{-z}}{2}$

9. $\sin^2 z + \cos^2 z = 1 \sin\ -z\ = -\sin z\ \cos\ -z\ = \cos z$

7.3.1 Logarithmic Functions

If $z = e^{w}$ then we write $w = \ln z$, called the natural logarithm of *z*. Thus the natural logarithm function is the inverse of the exponential function and can be defined by

$w = \ln z = \ln r + i\ \theta + 2\pi \cdot k$ where $k = 0, \pm 1, \pm 2, \pm 3, \ldots$

where $z = re^{i\theta} = re^{i\theta + 2\pi i}$.

So we have that w = ln z is multiple-valued, and in fact is infinitely many-valued. If we let k = 0 then $w = \ln z = \ln r + i\ \theta + 2\pi \cdot k\ = \ln r + i\theta$, where $0 \le \theta < 2\pi$ called the principal value or principal branch of $\ln z$.

To find the values of ln (1-i), let $z = e^{w}$, where $z = re^{i\theta} = r\ \cos\theta + i \sin\theta$ and so

$w = u + iv$ which implies $z = e^{u+iv} = e^{u}\ \cos v + i \sin v$.

We equate the real and imaginary parts:

1 $e^{u} \cos v = r \cos\theta$ 2 $e^{u} \cos v = r \sin\theta$

Squaring (1) and (2) and adding, we find $e^{2u} = r^2\ or\ e^{u} - r$ Thus, from (1) and (2),

$r \cos v = r \cos\theta$ $r \sin v = r \sin\theta$ *and so* $v = \theta + 2k\pi$

Hence we have $w = u + iv = \ln r + i\ \theta + 2k\pi$

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Example 13: Compute $\ln(1-i)$.

Since $1-i=\sqrt{2}e\ 1-i=\sqrt{2}e^{\frac{7\pi i}{4}+2k\pi i}$, we have

$$\ln(1-i)=\ln\sqrt{2}+\left(\frac{7\pi i}{4}+2k\pi i\right)=\frac{1}{2}\ln 2+\frac{7\pi i}{4}+2k\pi i$$

The principle value is $\frac{1}{2}\ln 2+\frac{7\pi i}{4}$ (let $k=0$).

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Practice Set

1. Write the following equations in a + jb form:-
   i) $x^2 + 4 = 0$
   ii) $x^2 + x + 1 = 0$
   iii) $x^2 + 6x + 11 = 0$
   iv) $x^3 - 1 = 0$

2. For the complex numbers z = 3 – j and w = 1 + 2j evaluate
   i) 2z – 3w
   ii) zw
   iii) $z^2 w \bar{z}^2$
   iv) $\frac{z}{w}$

3. Plot the following complex numbers on the Argand plane and put them into polar form.
   i) 1
   ii) j
   iii) – 3j
   iv) 1 – j
   v) 2 + j
   vi) – 3 – 2j
   viii) – 3 + 2j

4. Find the modulus of $e^{j\theta}$

5. Show that $(e^{j\theta})^{-1} = e^{-j\theta}$

6. Show that $\cos\theta = \frac{e^{jq} + e^{jq}}{2}$

   And $\sin\theta = \frac{e^{jq} e^{jq}}{2j}$

7. By conversion to polar form and use of De Moivres' theorem evaluate
   i) j^7
   ii) $(1 + j)^5$
   iii) $(\sqrt{3} - j)^{-4}$

8. Find in a + jb form and plot on Argand diagram:-
   i) The three values of $j^{1/3}$
   ii) The four values of $(1 + j)^{1/4}$

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9. Solve the following equations, writing the answer in z = a + jb form

i) $z^2 + 25 = 0$ ii) $z^2 + 4z + 5 = 0$

iii) $z^4 - 3z^2 - 4 = 0$ iv) $z^3 + z - 2 = 0$

v) $z^3 + 1 = 0$ vi) $z^2 + 2jz + 1 = 0$

Using egs (i) – (v) verify the result that the equations with real coefficients have real roots and/or complex roots occurring in conjugate pairs.

10. Find the real and imaginary parts of:

i) (1 – j) (1 + j)

ii) (3 – 4j) (1 + j)

iii) $(4 + 3j)^2$

iv) $(4 + 3j)^3$

11. Simplify the complex number $\frac{2j}{3+j} + \frac{1+j}{1j}$. Find the modulus and argument of the result.

12. State by inspection only (no arithmetic is necessary) whether each of the following numbers is purely real, purely imaginary or complex.

i) $\frac{4+j}{52j} - \frac{4j}{5+2j}$

ii) $\frac{j}{5+4j} - \frac{j}{54j}$

iii) $\left(\frac{1+2j}{2-3j}\right)^3 \left(\frac{1-2j}{2+3j}\right)^4$

13. Simplify i) $(1 + j\sqrt{3})^6 + (1 - j\sqrt{3})^6$ and ii) $(\sqrt{3} - j)^{15}$ by using De Moivre's theorem.

14. Simplify $\frac{\cos 3\theta - j\sin 3\theta^{-3} \cos 2\theta + j\sin 2\theta^{4}}{\cos 7\theta + j\sin 7\theta^{5} \cos 5\theta - j\sin 5\theta^{-4}}$

15. Determine all the roots in each case

i)Square roots of $1 + \sqrt{3}$ j

ii) Fourth roots of 1 – j

iii) Fourth roots of $\sqrt{3}$ + j

16. a) Express in the form $a + ib$,

(i) $(3 + i)^2$ (ii) $(2 + 4i)(3 + i)$.

b) The quadratic equation $z^2 - (2 + 4i)z + 8i - 6 = 0$ has roots z_1 and z_2.

i)Verify that $z_1 = 3 + i$ is a root of the equation.

ii)By considering the coefficients of the quadratic, write down the sum of the roots;

iii) Explain why z_1*, the complex conjugate of z_1, is **not** a root of the quadratic;

iv)Find the other root, z_2, in the form $a + ib$.

17. Find the square roots of 9 – 12i.

18. Find the modulus and argument of these complex numbers:

i) 2 + 7i

ii) 5 – 2i

iii) -4 + 3i

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19. The complex number 2 + 5i is one root of the equation $x^2 + px + q = 0$, where p and q are real.
i) Write down the second root of this equation.
ii) Find the values of p and q.

20. (i) Show that $z_1 = 2 + i$ is one of the roots of the equation $z^2 - 4z + 5 = 0$.Find the other root, z_2.

(ii) Show that $\frac{1}{z_1} + \frac{1}{z_2} = \frac{4}{5}$.

(iii) Show also that Im $(z_1^2 + z_2^2) = 0$ and find Re $(z_1^2 - z_2^2)$.

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Solutions: Practice questions

1. i) $x^2 + 4 = 0$

We have

$x^2 = -4$ or $x = \pm\sqrt{-4} = \pm 2\sqrt{-1} = \pm 2j$

ii) $x^2 + x + 1 = 0$

Using the formula for the solution of a quadratic we have

$$x = \frac{1 \pm \sqrt{3}}{2} = \frac{1 \pm \sqrt{3}\sqrt{1}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2} j$$

iii) $x^2 + 6x + 11 = 0$

$$x = \frac{6 \pm \sqrt{3644}}{2} = \frac{-6 \pm 2\sqrt{2}\sqrt{-1}}{2} = -3 \pm \sqrt{2}\ j$$

iv) $x^3 - 1 = 0$

Here we first factorize the cubic to give

$x^3 - 1 = (x - 1)(x^2 + x + 1) = 0$

This gives solutions, from ii)

$$x = 1,\ -\frac{1}{2} \pm \frac{\sqrt{3}}{2}\ j$$

2. i) $2z - 3w = 2(3 - j) - 3(1 + 2j) = 6 - 2j - 3 - 6j = 3 - 8j$

ii) $zw = (3 - j)(1 + 2j) = 3 - j + 6j - 2j^2 = 3 + 5j - 2(-1) = 5 + 5j$

on remembering that $j^2 = -1$

iii) Don't be so hasty with $z^2 w \bar{z}^2$ – look before you leap and notice that

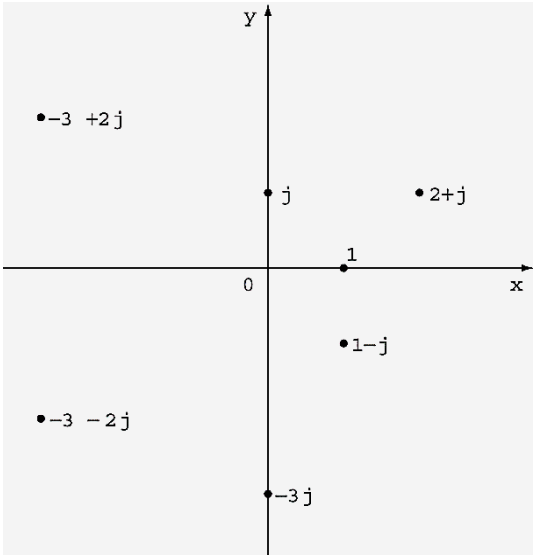
$z^2 w \bar{z}^2 = w(z\bar{z})^2 = (1 + 2j)\left(3^2 + (-1)^2\right) = 10(1 + 2j) = 10 + 20j$

iv) $$\frac{z}{w} = \frac{3 - j}{1 + 2j} = \frac{(3 - j)(1 - 2j)}{(1 + 2j)(1 - 2j)} = \frac{3 - 2 - j - 6j}{5} = \frac{1 - 7j}{5}$$

on multiplying top and bottom by the conjugate of $1 + 2j$

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3. Solution



i) 1 has modulus r = $\sqrt{1^2 + 0^2}$ = 1 and as it lies on the positive real axis the principal value of its argument is 0 radians, and so

$1 = 1 \angle(0)$

ii) |j| = 1 and j lies on the imaginary axis so its argument is $\theta = \frac{\pi}{2}$. So

$j = 1 \angle\left(\frac{\pi}{2}\right)$

iii) |– 3j| = 3 and the PV of the argument is $\theta = -\frac{\pi}{2}$, so

$-3j = 3 \angle\left(-\frac{\pi}{2}\right)$

iv) $|1 - j| = \sqrt{2}$ so

$1 - j = \sqrt{2}\left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}j\right)$

$= \sqrt{2}\left(\cos\left(\frac{\pi}{4}\right) - j\sin\left(\frac{\pi}{4}\right)\right)$

$= \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + j\sin\left(-\frac{\pi}{4}\right)\right)$

$= \sqrt{2}\ \angle\left(-\frac{\pi}{4}\right)$

So

$1 - j = \sqrt{2}\ \angle\left(-\frac{\pi}{4}\right)$

v) $2 + j = \sqrt{5}\left(\frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}}j\right) = \sqrt{5}\ \angle(\theta)$ where $\tan\theta = \frac{1}{2}$

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vi) $-3-2j=\sqrt{13}\left(-\frac{3}{\sqrt{13}}-\frac{2}{\sqrt{13}}j\right) = \sqrt{13}\ \angle(\theta)$ where $\tan\theta = \pi + \tan^{-1}\left(\frac{2}{3}\right)$

Here the complex number is in the third quadrant and we have chosen the angle in the range 0, 2π.

viii) $-3+2j=\sqrt{13}\left(-\frac{3}{\sqrt{13}}+\frac{2}{\sqrt{13}}j\right) = \sqrt{13}\ \angle(\theta)$ where $\tan\theta = \pi - \tan^{-1}\left(\frac{2}{3}\right)$

4. $|e^{j\theta}|\ \theta\ |\cos\theta + j\sin\theta| = \sqrt{\cos^2 q + \sin^2 q} = \sqrt{1} = 1$

5. $(e^{j\theta})^{-1} = (\cos\theta + j\sin\theta)^{-1} = \frac{1}{\cos q + j\sin q}$

$= \frac{\cos q - j\sin q}{\cos^2 q + \sin^2 q} = \cos\theta - j\sin\theta = \cos(-\theta) + j\sin(-\theta) = e^{-j\theta}$

6. We have

$e^{j\theta} = \cos\theta + j\sin\theta$

$e^{-j\theta} = \cos\theta - j\sin\theta$

Adding these identities gives

$2\cos\theta = e^{j\theta} + e^{-j\theta}$ so $\cos\theta = \frac{e^{jq} + e^{-jq}}{2}$

Subtracting the results gives

$2j\sin\theta = e^{j\theta} - e^{-j\theta}$ so $\sin\theta = \frac{e^{jq} - e^{-jq}}{2j}$

7. i) $j^7 = \left(\angle\left(\frac{\pi}{2}\right)\right)^7 = \angle\left(\frac{7\pi}{2}\right) = \angle\left(\frac{3\pi}{2}\right) = -j$

ii) $(1+j)^5 = \left(\sqrt{2}\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}j\right)\right)^5 = (\sqrt{2})^5\left(\angle\left(\frac{\pi}{4}\right)\right)^5 = 4\sqrt{2}\ \angle\left(\frac{5\pi}{4}\right)$

$= -4\sqrt{2}\left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}j\right) = -4(1+j)$

iii) First, we have $\sqrt{3} - j = 2\left(\frac{\sqrt{3}}{2}-\frac{1}{2}j\right) = 2\left(\cos\left(\frac{\pi}{6}\right) - j\sin\left(\frac{\pi}{6}\right)\right) = 2\angle\left(-\frac{\pi}{6}\right)$

So

$(\sqrt{3}-j)^{-4} = \left(2\angle\left(-\frac{\pi}{6}\right)\right)^{-4} = 2^{-4}\angle\left(\frac{2\pi}{3}\right) = 2^{-4}\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}j\right)$

8. i) $j^{1/3} = \left(\angle\left(\frac{p}{2}\right)\right)^{\frac{1}{3}} = \left(\angle\left(\frac{p}{2}+2kp\right)\right)^{\frac{1}{3}}$

$= \angle\left(\frac{p}{6}+\frac{2kp}{3}\right)$ for k = 0, 1, 2

k = 0 gives

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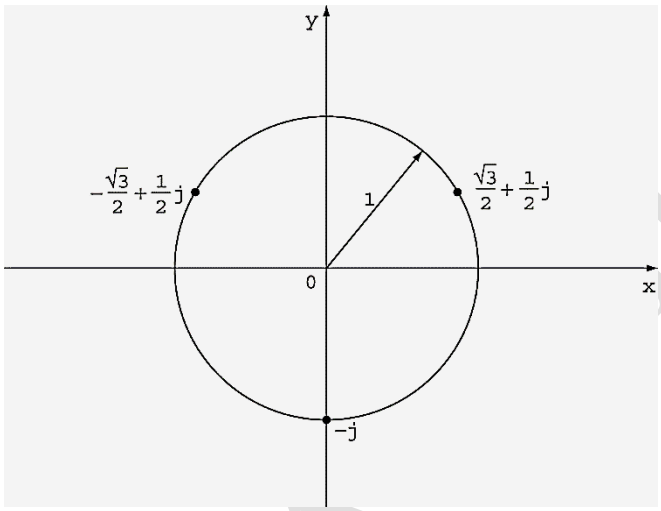
$= \angle\left(\frac{p}{6}\right) = \cos\left(\frac{p}{6}\right) + j \sin\left(\frac{p}{6}\right) = \frac{\sqrt{3}}{2} + \frac{1}{2} j$

k = 1 gives

$= \angle\left(\frac{5p}{6}\right) = \cos\left(\frac{5p}{6}\right) + j \sin\left(\frac{5p}{6}\right) = -\frac{\sqrt{3}}{2} + \frac{1}{2} j$

k = 2 gives

$= \angle\left(\frac{3p}{2}\right) = -j$



ii) $(1 + j)^{1/4} = \left(\sqrt{2}\left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} j\right)\right)^{\frac{1}{4}}$

$= 2^{\frac{1}{8}} \left(\angle\left(\frac{p}{4}\right)\right)^{\frac{1}{4}} = 2^{\frac{1}{8}} \left(\angle\left(\frac{p}{4} + 2kp\right)\right)^{\frac{1}{4}}$

$= 2^{\frac{1}{8}} \angle\left(\frac{p}{16} + \frac{kp}{2}\right)$ for k = 0, 1, 2, 3

k = 0 gives

$= 2^{\frac{1}{8}} \angle\left(\frac{p}{16}\right) = 2^{1/8}\left(\cos\left(\frac{p}{16}\right) + j \sin\left(\frac{p}{16}\right)\right)$

k = 1 gives

$= 2^{\frac{1}{8}} \angle\left(\frac{p}{16} + \frac{p}{2}\right) = 2^{\frac{1}{8}} \angle\left(\frac{9p}{16}\right)$

$= 2^{1/8}\left(\cos\left(\frac{9p}{16}\right) + j \sin\left(\frac{9p}{16}\right)\right)$

k = 2 gives

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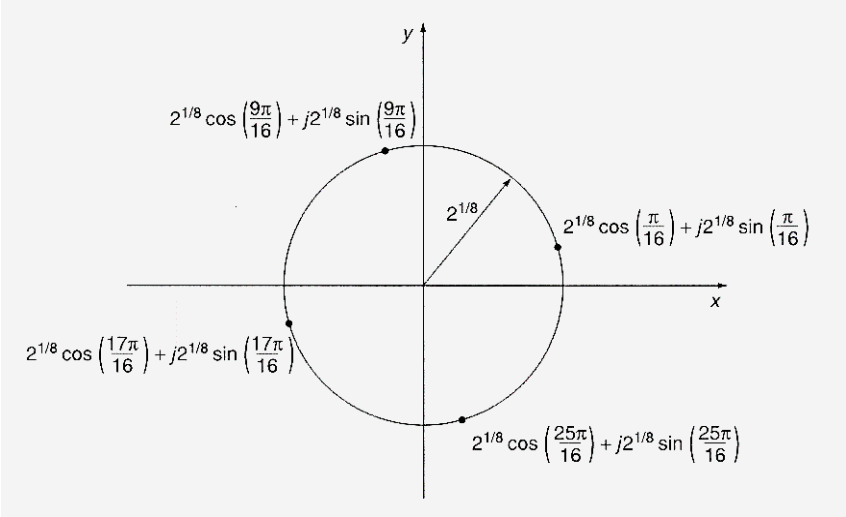
$$2^{\frac{1}{8}} \angle\left(\frac{p}{16} + p\right) = 2^{\frac{1}{8}} \angle\left(\frac{17p}{16}\right)$$

$$= 2^{1/8}\left(\cos\left(\frac{17p}{16}\right) + j \sin\left(\frac{17p}{16}\right)\right)$$

k = 3 gives

$$2^{\frac{1}{8}} \angle\left(\frac{p}{16} + \frac{3p}{2}\right) = 2^{\frac{1}{8}} \angle\left(\frac{25p}{16}\right)$$

$$= 2^{1/8}\left(\cos\left(\frac{25p}{16}\right) + j \sin\left(\frac{25p}{16}\right)\right)$$



9. i) $z^2 + 25 = 0$ gives $z^2 = -25$ so

$$z = \pm\sqrt{-25} = \pm\sqrt{-5^2} = \pm 5\sqrt{-1} = \pm 5j$$

ii) $z^2 + 4z + 5 = 0$ gives, using the quadratic formula,

$$z = \frac{-4 \pm \sqrt{4^2 - 4¥5}}{2} = \frac{-4 \pm \sqrt{16 - 20}}{2}$$

$$\frac{-4 \pm \sqrt{-4}}{2} = \frac{-4 \pm 2j}{2} = -2 \square j$$

iii) $z^4 - 3z^2 - 4 = 0$ is really a quadratic equation in disguise, because if we put $u = z^2$ then we get

$$z^4 - 3z^2 - 4 = u^2 - 3u - 4 = (u + 1)(u - 4) = 0$$

So $u = -1$ or $z^2 = -1$ and hence $z = \pm j$ or $u = 4 = z^2$ so $z = \pm 2$

iv) To solve $z^3 + z - 2 = 0$ we first factorize it. By the factor theorem, the fact that $z = 1$ is (by trial) a root tells us that $z - 1$ is a factor, so we can write

$$z^3 + z - 2 = (z - 1)(z^2 + az + b)$$

Equating coefficients gives $b = 2$ and $-a + b = 1$, so $a = 1$ and hence

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$z^3 + z - 2 = (z - 1)(z^2 + z + 2) = 0$

Thus, $z = 1$ or $z^2 + z + 2 = 0$ i.e.

$$z = \frac{-1 \pm \sqrt{1 - 8}}{2} = \frac{-1 \pm \sqrt{7}j}{2}$$

v) Using the factorization of a sum of cubes we have

$z^3 + 1 = (z + 1)(z^2 - z + 1) = 0$

and so

$$z = -1 \text{ or } z = \frac{1 \pm \sqrt{1 - 4}}{2} = \frac{-1 \pm \sqrt{3}j}{2}$$

vi) Even though the equation $z^2 + 2jz + 1 = 0$ has complex coefficients we can still use the usual quadratic formula to obtain the solution:

$$z = \frac{-2j \pm \sqrt{(2j)^2 - 4}}{2} = \frac{-2j \pm \sqrt{-8}}{2}$$

$$\frac{-2j \pm 2\sqrt{2}\,j}{2} = (-1 \pm \sqrt{2})j$$

The roots of all equations i) to v) are either real or occur in conjugate pairs – for example $-2 + j$ and $-2 - j$. The roots of vi) are not conjugate pairs because the coefficients of the equation are not all real.

10. i) $(1 - j)(1 + j) = 1 - j^2 = 1 - (-1) = 2$

So real part = 2, imaginary part = 0

ii) $(3 - 4j)(1 + j) = 3 + 3j - 4j - 4j^2 = 3 - j + 4 = 7 - j$

So real part = 7, imaginary part = –1

iii) $(4 + 3j)^2 = 4^2 + 24j + (3j)^2 = 16 + 24j - 9 = 7 + 24j$

So real part = 7, imaginary part = 24

iv) By the binomial theorem

$(4 + 3j)^3 = 4^3 + 3.\,4^2.\,3j + 3.4.\,(3j)^2 + (3j)^3$

$= 64 + 6\,.\,16j - 4.\,27 - 27j = -44 + 69j$

So real part = – 44, imaginary part = 69

11. $$z = \frac{2 - j}{3 + j} + \frac{1 + j}{1 - j} = \frac{(2 - j)(1 - j) + (1 + j)(3 + j)}{(3 + j)(1 - j)}$$

$$= \frac{2 - 3j - 1 + 3 + 4j - 1}{3 - 2j + 1} = \frac{3 + j}{4 - 2j} = \frac{3 + j}{4 - 2j}\frac{4 + 2j}{4 + 2j} = \frac{12 - 2 + 10j}{16 + 4}$$

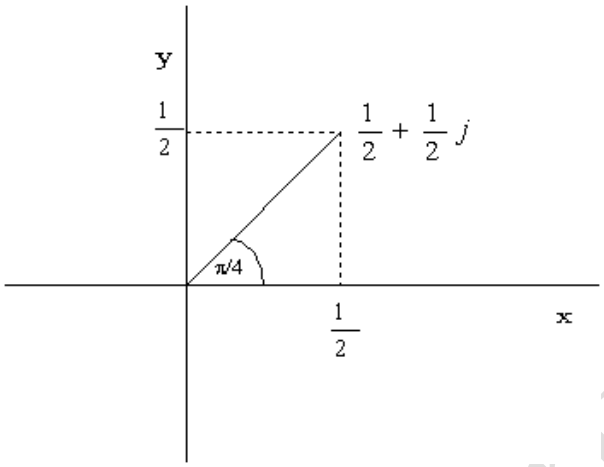
$$= \frac{10 + 10j}{20} = \frac{1}{2} + \frac{1}{2}\,j$$

For the modulus we have

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$$|z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

The argument is seen to be $\frac{p}{4}$ from the figure.



12. i) If $z = \frac{4+j}{5-2j} - \frac{4-j}{5+2j}$ then we have

$$\bar{z} = \frac{4-j}{5+2j} - \frac{4+j}{5-2j} = -\left(\frac{4+j}{5-2j} - \frac{4-j}{5+2j}\right) = -z$$

So z is pure imaginary.

ii) If $z = \frac{j}{5+4j} - \frac{j}{5-4j}$ then

$$\bar{z} = \frac{-j}{5-4j} - \frac{-j}{5+4j} = \frac{j}{5+4j} - \frac{j}{5-4j} = z$$

So z is real.

iii) If $z = \left(\frac{1+2j}{2-3j}\right)^3 \left(\frac{1-2j}{2+3j}\right)^4$ then

$$\bar{z} = \left(\frac{1-2j}{2+3j}\right)^3 \left(\frac{1+2j}{2-3j}\right)^4$$

This has no simple relation to z and is certainly not equal to ± z, so in this case z is neither real nor pure imaginary.

13. i) First put $1 + j\sqrt{3}$ and $1 - j\sqrt{3}$ into polar form:

$$1 + j\sqrt{3} = 2\left(\frac{1}{2} + \frac{\sqrt{3}}{2}j\right) = 2\ \angle\left(\frac{\pi}{3}\right)$$

and similarly:

$$1 - j\sqrt{3} = 2\left(\frac{1}{2} - \frac{\sqrt{3}}{2}j\right) = 2\ \angle\left(-\frac{\pi}{3}\right)$$

So

$$\left(1 + j\sqrt{3}\right)^6 + \left(1 - j\sqrt{3}\right)^6 = \left(2\ \angle\left(\frac{\pi}{3}\right)\right)^6 + \left(2\ \angle\left(-\frac{\pi}{3}\right)\right)^6$$

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$= 2^6 \angle(2\pi) + 2^6 \angle(-2\pi) = 2^6 + 2^6 = 2^7$

ii) First put $\sqrt{3} - j$ into polar form:

$\sqrt{3} - j = 2\left(\frac{\sqrt{3}}{2} - \frac{1}{2}j\right) = 2\left(\cos\left(\frac{\pi}{6}\right) - j\sin\left(\frac{\pi}{6}\right)\right)$

$= 2\left(\cos\left(-\frac{\pi}{6}\right) + j\sin\left(-\frac{\pi}{6}\right)\right) = 2\angle\left(-\frac{\pi}{6}\right)$

So

$(\sqrt{3} - j)^{15} = \left(2\angle\left(-\frac{\pi}{6}\right)\right)^{15} = 2^{15}\angle\left(-\frac{15\pi}{6}\right)$

$= 2^{15}\angle\left(-\frac{5\pi}{2}\right) = 2^{15}\left(\cos\left(-\frac{5\pi}{2}\right) + j\sin\left(-\frac{5\pi}{2}\right)\right)$

$= 2^{15}\left(-j\sin\left(\frac{5\pi}{2}\right)\right) = -2^{15}j$

14. $\frac{\cos 3\theta - j\sin 3\theta^{-3}\ \cos 2\theta + j\sin 2\theta^{4}}{\cos 7\theta + j\sin 7\theta^{5}\ \cos 5\theta - j\sin 5\theta^{-4}}$

$= \frac{\cos 9\theta + j\sin 9\theta\ \ \cos 8\theta + j\sin 8\theta}{\cos 35\theta + j\sin 35\theta\ \ \cos 20\theta + j\sin 20\theta}$

$= \frac{\cos 17\theta + j\sin 17\theta}{\cos 55\theta + j\sin 55\theta}$

$= \cos(-38\theta) + j\sin(-38\theta)$

$= \cos(38\theta) - j\sin(38\theta)$

15. i) $(1 + \sqrt{3}\ j)^{1/2} = \left(2\angle\left(\frac{\pi}{3}\right)\right)^{1/2} = 2^{1/2}\left(\angle\left(\frac{\pi}{3} + 2k\pi\right)\right)^{1/2}$

$= \sqrt{2}\ \angle\left(\frac{\pi}{6} + k\pi\right)$ for k = 0, 1

For k = 0 we get

$= \sqrt{2}\angle\left(\frac{\pi}{6}\right) = \sqrt{2}\left(\cos\left(\frac{\pi}{6}\right) + j\sin\left(\frac{\pi}{6}\right)\right)$

$= \sqrt{2}\left(\frac{\sqrt{3}}{2} + \frac{1}{2}j\right)$

For k = 1 we get

$= \sqrt{2}\angle\left(\frac{7\pi}{6}\right) = \sqrt{2}\left(\cos\left(\frac{7\pi}{6}\right) + j\sin\left(\frac{7\pi}{6}\right)\right)$

$= \sqrt{2}\left(-\frac{\sqrt{3}}{2} - \frac{1}{2}j\right)$

We can summarise the two answers as:

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$\pm\sqrt{2}\left(\frac{\sqrt{3}}{2}+\frac{1}{2}j\right)$

ii) $(1-j)^{1/4}=\left(\sqrt{2}\angle\left(-\frac{\pi}{4}\right)\right)^{1/4}=\ 2^{1/8}\left(\angle\left(-\frac{\pi}{4}+2k\pi\right)\right)^{1/4}$

$=\ 2^{1/8}\angle\ \ \text{for k = 0, 1, 2, 3}$

For k = 0 we get

$=\ 2^{1/8}\angle\left(-\frac{\pi}{16}\right)\ =2^{1/8}\left(\cos\left(\frac{\pi}{16}\right)-j\sin\left(\frac{\pi}{16}\right)\right)$

For k = 1 we get

$=\ 2^{1/8}\angle\left(\frac{7\pi}{16}\right)\ =\ 2^{1/8}\left(\cos\left(\frac{7\pi}{16}\right)+j\sin\left(\frac{7\pi}{16}\right)\right)$

For k = 2 we get

$=\ 2^{1/8}\angle\left(\frac{15\pi}{16}\right)\ =\ 2^{1/8}\left(\cos\left(\frac{15\pi}{16}\right)+j\sin\left(\frac{15\pi}{16}\right)\right)$

For k = 3 we get

$=\ 2^{1/8}\angle\left(\frac{23\pi}{16}\right)\ =\ 2^{1/8}\left(\cos\left(\frac{23\pi}{16}\right)+j\sin\left(\frac{23\pi}{16}\right)\right)$

$=2^{1/8}\left(\cos\left(-\frac{9\pi}{16}\right)+j\sin\left(-\frac{9\pi}{16}\right)\right)$

$=2^{1/8}$

iii) $(\sqrt{3}+j)^{1/4}=\left(2\angle\left(\frac{\pi}{6}\right)\right)^{1/4}=\ 2^{1/4}\left(\angle\left(\frac{\pi}{6}+2k\pi\right)\right)^{1/4}$

$=\ 2^{1/4}\angle\left(\frac{\pi}{24}+\frac{k\pi}{2}\right)\ \ \text{for k = 0, 1, 2, 3}$

For k = 0 we get

$=\ 2^{1/4}\angle\left(\frac{\pi}{24}\right)\ =2^{1/4}\left(\cos\left(\frac{\pi}{24}\right)+j\sin\left(\frac{\pi}{24}\right)\right)$

For k = 1 we get

$=\ 2^{1/4}\angle\left(\frac{13\pi}{24}\right)\ =\ 2^{1/4}\left(\cos\left(\frac{13\pi}{24}\right)j\sin\left(\frac{13\pi}{24}\right)\right)$

For k = 2 we get

$=\ 2^{1/4}\angle\left(\frac{25\pi}{24}\right)\ =\ 2^{1/4}\angle\left(-\frac{23\pi}{24}\right)\ =2^{1/4}\left(\cos\left(-\frac{23\pi}{24}\right)+j\sin\left(-\frac{23\pi}{24}\right)\right)$

$=2^{1/4}\left(\cos\left(\frac{23\pi}{24}\right)-j\sin\left(\frac{23\pi}{24}\right)\right)$

For k = 3 we get

$=\ 2^{1/4}\angle\left(\frac{37\pi}{24}\right)\ =2^{1/4}\angle\left(-\frac{11\pi}{24}\right)$

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$= \ 2^{1/4}\left(\cos\left(-\frac{11\pi}{24}\right) + \text{j}\sin\left(-\frac{11\pi}{24}\right)\right)$

$= 2^{1/4}\left(\cos\left(\frac{11\pi}{24}\right) - \left(\frac{11\pi}{24}\right)\right)$

16. a)

(i) $(3 + \text{i})^2 = (3 + i)(3 + i) = 9 + 3i + 3i + i^2$

$= 9 + 6i - 1 = 8 + 6i$ Remember: $\text{i}^2 = -1$

(ii) $(2 + 4i)(3 + i) = 6 + 2i + 12i + 4i^2$
$= 6 + 14i - 4 = 2 + 14i$

b) (i)We substitute $z = 3 + i$ into the expression $z^2 - (2 + 4i)z + 8i - 6$ to check that it gives an answer of 0.

$(3 + i)^2 - (2 + 4i)(3 + i) + 8i - 6$

$= 8 + 6i - (2 + 14i) + 8i - 6$ (using the results from (a))

$= 8 + 6i - 2 - 14i + 8i - 6$

$= 0 + 0i = 0$ (as required)

(ii) In a quadratic equation, the sum of roots is given by the expression $-b/a$.

So here the sum of the roots is $(2 + 4i)$ (since $b = -(2 + 4i)$ and $a = 1$)

(iii) In a quadratic equation with real coefficients, any complex roots form a conjugate pair. However, this quadratic does not have real coefficients so the roots are not complex conjugates of each other.

(iv) From (ii), $z_1 + z_2 = 2 + 4i$
So $3 + i + z_2 = 2 + 4i$

i.e. $z_2 = 2 + 4i - (3 + i) = -1 + 3i$.

17. Let $a + bi$ be a square root of $9 - 12i$, where a and b are real numbers.

Then $(a + bi)^2 = 9 - 12i$.

So:

$(a + bi)(a + bi) = 9 - 12i$

i.e. $a^2 + abi + abi + b^2i^2 = 9 - 12\text{i}$

i.e. $a^2 - b^2 + 2abi = 9 - 12i$.

Comparing real and imaginary parts, we get two equations:

$a^2 - b^2 = 9$ ①

and $ab = -6$ ②

From equation (2), we get. $b = \frac{-6}{a}$

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Substituting this into (1) gives: $a^2 - \left(\frac{-6}{a}\right)^2 = 9$

i.e. $a^2 - \frac{36}{a^2} = 9$

i.e. $a^4 - 36 = 9a^2$

or $a^4 - 9a^2 - 36 = 0$

This is a quadratic equation in a^2. It can be factorised:

$(a^2 - 12)(a^2 + 3) = 0$

So $a^2 = 12$ or $a^2 = -3$.

As a is real, we must have $a = \pm\sqrt{12} = \pm 2\sqrt{3}$.

If $a = 2\sqrt{3}, then\, b = \frac{-6}{a} = \frac{-6}{2\sqrt{3}} = -\sqrt{3}$.

$If\ a = a - 2\sqrt{3}, then\, b = \frac{-6}{a} = \frac{-6}{-2\sqrt{3}} = \sqrt{3}$.

Therefore the square roots of $9 - 12i$ are $2\sqrt{3} - \sqrt{3i}$ and $-2\sqrt{3} + \sqrt{i}$.

Note: All questions about finding the square root of a complex number can be solved using the same method.

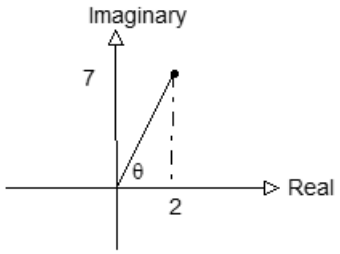
18. It helps to sketch an Argand diagram in each case:

a) $|\, 2 + 7i\, | = \sqrt{2^2 + 7^2} = \sqrt{53}$

arg(2 + 7i) is angle θ in the diagram:

$\tan\theta = \frac{7}{2}$

i.e. θ = 1.29 radians



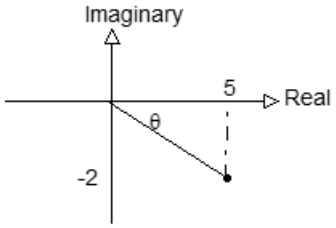
b) $|\, 5 - 2i\, | = \sqrt{5^2 + -2^{\,2}} = \sqrt{29}$

arg(5 - 2i) is shown by angle θ in the diagram.

$\tan\theta = \frac{2}{5} \Rightarrow \theta = 0.381$

i.e. θ = -0.381 radians (it is negative as the number is below the axis).

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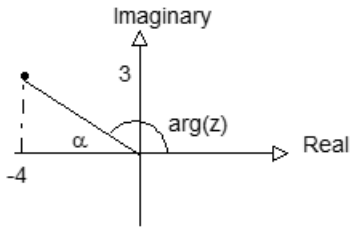
c) $|-4+3i| = \sqrt{-4^{2}+3^{2}} = 5$

arg(-4 + 3i) is the angle shown on the diagram.

To find it, it is easiest to first find angle α:

$\tan(\alpha) = \frac{3}{4} \quad \Rightarrow \quad \alpha = 0.644$

Therefore, arg(-4 + 3i) = π – 0.644 = 2.50 radians.



19. a) As the quadratic equation has real coefficients, the second root must be the complex conjugate of the first, i.e. 2 – 5i.

b) We know that the sum of the roots is given by $-b/a = -p$.

But the sum of the roots is (2 + 5i) + (2 – 5i) = 4.

Therefore $-p = 4$ i.e. $p = -4$.

The product of the roots is $c/a = q$.

But the product of the roots is (2 + 5i)(2 – 5i) = 4 – 10i + 10i + 25 = 29.

So $q = 29$.

20. (i) $z_1^2 - 4x + 5 = (2+i)^2 - 4(2+i) + 5$

$= 4 + 4i - 1 - 8 - 4i + 5$

$= 0$

The other root, $z_2 = z_1^* = 2 - i$

(ii) $\frac{1}{z_1} + \frac{1}{z_2} = \frac{1}{2+i} + \frac{1}{2-i}$

$= \frac{2-i+2+i}{(2+i)(2-i)}$

$= \frac{4}{4+1}$

$= \frac{4}{5}$

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(iii) $z_1^2 + z_2^2 = (2+i)^2 + (2-i)^2$

$= 4+4i-1+4-4i-1$

$= 6$

$lm\ z_1^2 + z_2^2 = 0$

$z_1^2 - z_2^2 = (2+i)^2 - (2-i)^2$

$= 4+4i-1-4+4i+1$

$= 8i$

$\text{Re}\ z_1^2 - z_2^2 = 0$

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7.3.2 Analytic function

In mathematics, an **analytic function** is a function that is locally given by a convergent power series. Analytic functions can be thought of as a bridge between polynomials and general functions. There exist both **real analytic functions** and **complex analytic functions**, categories that are similar in some ways, but different in others. Functions of each type are infinitely differentiable, but complex analytic functions exhibit properties that do not hold generally for real analytic functions. A function is analytic if it is equal to its Taylor series in some neighborhood.

Formally, a function *f* is *real analytic* on an open set *D* in the real line if for any x_0 in *D* one can write

$$f(x) = \sum_{n=0}^{\infty} a_n (x - x_0)^n = a_0 + a_1 (x - x_0) + a_2 (x - x_0)^2 + a_3 (x - x_0)^3 + \ldots$$

in which the coefficients a_0, a_1, ... are real numbers and the series is convergent for *x* in a neighborhood of x_0.

Alternatively, an analytic function is an infinitely differentiable function such that the Taylor series at any point x_0 in its domain

$$T(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n$$

is convergent for *x* close enough to x_0 and its value equals $f(x)$.

The definition of a *complex analytic function* is obtained by replacing everywhere above "real" with "complex", and "real line" with "complex plane".

Most special functions are analytic (at least in some range of the complex plane). Typical examples of analytic functions are:

- Any polynomial (real or complex) is an analytic function. This is because if a polynomial has degree *n*, any terms of degree larger than *n* in its Taylor series expansion will vanish, and so this series will be trivially convergent.
- The exponential function is analytic. Any Taylor series for this function converges not only for *x* close enough to x_0 (as in the definition) but for all values of *x* (real or complex).
- The trigonometric functions, logarithm, and the power functions are analytic on any open set of their domain.
- The absolute value function when defined on the set of real numbers or complex numbers is not analytic because it is not differentiable at 0. Piecewise defined functions (functions given by different formulas in different regions) are in general not analytic.
- The complex conjugate function, $z \to \bar{z}$ is not complex analytic, although its restriction to the real line is real analytic.

Properties of analytic functions

- The sums, products, and compositions of analytic functions are analytic.
- The reciprocal of an analytic function that is nowhere zero is analytic, as is the inverse of an invertible analytic function whose derivative is nowhere zero. (See also the Lagrange inversion theorem.)

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Analyticity and differentiability

As noted above, any analytic function (real or complex) is infinitely differentiable (also known as smooth, or C^∞). (Note that this differentiability is in the sense of real variables; compare complex derivatives below.) There exist smooth real functions which are not analytic: see the following example. In fact there are many such functions, and the space of real analytic functions is a proper subspace of the space of smooth functions.

The situation is quite different when one considers complex analytic functions and complex derivatives. It can be proved that any complex function differentiable (in the complex sense) in an open set is analytic. Consequently, in complex analysis, the term *analytic function* is synonymous with *holomorphic function*.

- Any analytic function is smooth, that is, infinitely differentiable. The converse is not true; in fact, in a certain sense, the analytic functions are rather sparse compared to the infinitely differentiable functions.
- For any open set $\Omega \subseteq \mathbf{C}$, the set $A(\Omega)$ of all bounded, analytic functions $u : \Omega \to \mathbf{C}$ is a Banach space with respect to the supremum norm. The fact that uniform limits of analytic functions are analytic is an easy consequence of Morera's theorem.

A polynomial cannot be zero at too many points unless it is the zero polynomial (more precisely, the number of zeros is at most the degree of the polynomial). A similar but weaker statement holds for analytic functions. If the set of zeros of an analytic function *f* has an accumulation point inside its domain, then *f* is zero everywhere on the connected component containing the accumulation point.

7.4 The Cauchy-Riemann Equations and Complex Differentiation

Suppose that we consider $f(z) = z^2$ and substitute into this z = x+iy. We can therefore write this function again in the following way:

$f(z) = z^2 = F(x,y) = (x+iy)^2 = (x^2 - y^2) + i2xy = u(x,y) + iv(x,y)$

where $u(x,y) = (x^2 - y^2)$ and where $v(x,y) = 2xy$. Now since z = x+iy, we know that

$$x = \frac{z+\overline{z}}{2} \quad \text{and} \quad y = \frac{z-\overline{z}}{2i}$$

from which it follows that $\frac{\partial x}{\partial z} = \frac{1}{2}$ and $\frac{\partial y}{\partial z} = \frac{1}{2i}$. Now consider the complex derivative f '(z).

$$f'(z) = \frac{\partial F}{\partial x}\frac{\partial x}{\partial z} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial z} = (2x - 2yi)(\frac{1}{2}) + (-2y + 2xi)(\frac{1}{2i})$$

This of course reduces to $f'(z) = 2z$ and the result agrees with the derivative computed in the previous section using limits.

Now suppose that F(x,y) = u(x,y) + iv(x,y) is *any* differentiable complex function. What must be true about the functions u and v ? This is the subject of the Cauchy - Riemann equations.

First, suppose that z changes by x changing alone. Then, assume that z changes by y changing alone. This would give us two expressions for the derivative of

f(z) = F(x,y).

The first (*holding y constant*) can be written as

$$f'(z)|_{y\ constant} = \frac{\partial F}{\partial x}\frac{\partial x}{\partial z} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial z} + i\frac{\partial v}{\partial x}\frac{\partial x}{\partial z} = \frac{1}{2}\{\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\}$$

while the second (*holding x constant*) can be written as

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$$f'(z)|_{x\ constant} = \frac{\partial F}{\partial y}\frac{\partial y}{\partial z} = \frac{\partial u}{\partial y}\frac{\partial y}{\partial z} + i\frac{\partial v}{\partial y}\frac{\partial y}{\partial z} = \frac{1}{2}\{-i\frac{\partial u}{\partial y} + \frac{\partial v}{\partial y}\}$$

Now, the derivative of f(z) cannot depend on which way that z is changing (either by x changing alone or alternatively by y changing alone) and so the two expressions for $f'(z)$ must be equal if the derivative exists. This implies that

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and that } -\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}$$

These two equalities are known as the Cauchy-Riemann Equations or Cauchy-Riemann conditions. These Cauchy-Riemann conditions are necessary for the existence of a derivative, that is, if $f'(x)$ exists, the C-R conditions must hold.

Conversely, if the C-R conditions are satisfied and the partial derivatives of u(x,y) and v(x,y) are continuous, $f'(z)$ exists.

7.4.1 Cauchy's integral theorem

If a function f(z) is analytical (therefore single-valued) [and its partial derivatives are continuous] through some simply connected region **R**, for every closed path C (Fig. 1) in **R**,

$$\oint_c f\ z\ dz = 0$$

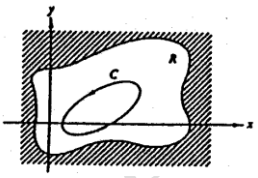


Fig. 1

Stokes's theorem proof

Proof: (under relatively restrictive condition: the partial derivative of u, v are continuous, which are actually not required but usually satisfied in physical problems)

$$\oint_c f\ z\ dz = \oint_c udx - vdy + i\oint_c vdx + udy$$

These two line integrals can be converted to surface integrals by Stokes's theorem $\left(\oint_c \underline{A}\cdot d\underline{l} = \int_s \nabla\times\underline{A}\cdot d\underline{s}\right)$.

Using $\underline{A} = A_x\hat{x} + A_y\hat{y}$ and $ds = dxdy\hat{z}$,

We have

$$\oint_c A_x dx + A_y dy = \oint_c \underline{A}\cdot d\underline{l} = \int_s \nabla\times\underline{A}\cdot d\underline{s}$$

$$= \int\left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y}\right)dxdy$$

If we let u = A_x, and v = -A_y, then

$$\oint_c udx - vdy = -\int\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)dxdy$$

=0 [since C-R conditions $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$]

Setting u = A_y and v = A_x, we have

$$\oint vdx + udy = \int\left(\frac{\partial u}{\partial x}-\frac{\partial v}{\partial y}\right)dxdy = 0$$

$$\therefore \qquad \oint f\ z\ dz = 0$$

The consequence of the theorem is that for analytic functions the line integral is a function only of its end points, independent of the path of integration,

$$\int_{z_1}^{z_2} f\ z\ dz = F\ z_2\ - F\ z_1\ = -\int_{z_2}^{z_1} f\ z\ dz$$

Multiply connected regions

The original statement of our theorem demanded a simply connected region. This restriction may easily be relaxed by the creation of a barrier, a contour line. Consider the multiply connected region of Fig.2 In which f(z) is not defined for the interior R′

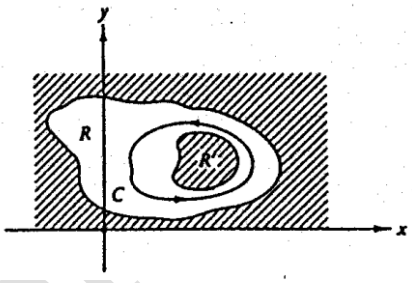


Fig. 2

Cauchy's integral theorem is not valid for the contour C, but we can construct a C′ for which the theorem holds. If line segments DE and GA arbitrarily close together, then

$$\int_G^A f\ z\ dz = -\int_D^E f\ z\ dz$$

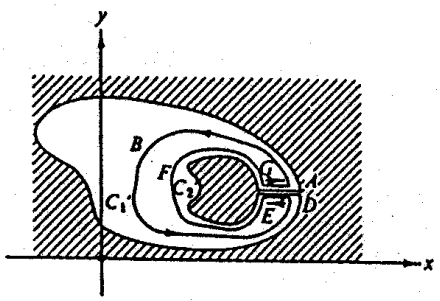


Fig. 3

Now let ABD→C_1' and EFG→$-C_2'$, we obtain

$$\oint_{C'_{ABDEFGA}} f\ z\ dz = \left[\int_{ABD} + \int_{DE} + \int_{GA} + \int_{EFG}\right] f\ z\ dz$$

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$$= \left[\int_{ABD} + \int_{EFG} \right] f\ z\ dz = 0$$

$$\Downarrow$$

$$\oint_{C_1'} f\ z\ dz = \oint_{C_2'} f\ z\ dz$$

Cauchy's Integral Formula

If f(z) is analytic on and within a closed contour C then

$$\oint_C \frac{f\ z\ dz}{z - z_0} = 2\pi i f\ z_0$$

In which z_0 is some point in the interior region bounded by C. Note that here $z\text{-}z_0 \neq 0$ and the integral is well defined.

Although f(z) is assumed analytic, the integrand $(f(z)/z\text{-}z_0)$ is not analytic at $z=z_0$ unless $f(z_0)=0$. If the contour is deformed as in Fig.4 Cauchy's integral theorem applies.

So we have

$$\oint_C \frac{f\ z\ dz}{z - z_0} - \oint_{C_2} \frac{f\ z}{z - z_0} dz = 0$$

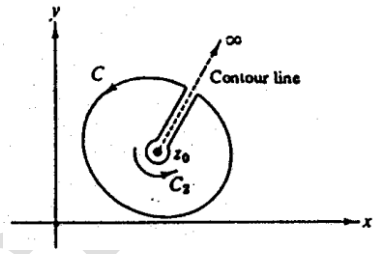


Fig. 4

Let $z - z_0 = re^{i\theta}$, here r is small and will eventually be made to approach zero

$$\therefore \quad \oint_{C_2} \frac{f\ z\ dz}{z - z_0} dz = \oint_{C_2} \frac{f\ z_0 + re^{i\theta}}{re^{i\theta}} rie^{i\theta} d\theta$$

$$(r \to 0) \quad = if\ z_0 \oint_{C_2} d\theta = 2\pi i f\ z_0$$

Here is a remarkable result. The value of an analytic function is given at an interior point at $z=z_0$ once the values on the boundary C are specified.

What happens if z_0 is exterior to C?

In this case the entire integral is analytic on and within C, so the integral vanishes.

$$\frac{1}{2\pi i} \oint_C \frac{f\ z\ dz}{z - z_0} = \begin{cases} f\ z_0\ , & z_0 \text{interior} \\ 0, & z_0 \text{ exterior} \end{cases}$$

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•*Derivatives*

Cauchy's integral formula may be used to obtain an expression for the derivation of f(z)

$$f\ z_0 = \frac{d}{dz_0}\left(\frac{1}{2\pi i}\oint \frac{f\ z\ dz}{z - z_0}\right)$$

$$= \frac{1}{2\pi i}\oint f\ z\ dz \frac{d}{dz_0}\left(\frac{1}{z - z_0}\right) = \frac{1}{2\pi i}\oint \frac{f\ z\ dz}{z - z_0\ ^2}$$

Moreover, for the n-th order of derivative

$$f^{\ n}\ z_0 = \frac{n!}{2\pi i}\oint \frac{f\ z\ dz}{z - z_0\ ^{n+1}}$$

We now see that, the requirement that f(z) be analytic not only guarantees a first derivative but derivatives of all orders as well! The derivatives of f(z) are automatically analytic. Here, it is worth to indicate that the converse of Cauchy's integral theorem holds as well.

Examples:

Evaluate integral using Cauchy Integral Formula

$$\int_C \frac{z^2 - 1}{z^2 + 1} dz\ in\ counter - clockwise\ direction\ around\ the\ countour$$

i) $|z| = 0.5$ ii) $|z + i| = 1$ iii) $|z - i| = 1$ iv) $|z - 2i| = 2$

Firstly denominator can be expressed as product of linear functions (z+i) and (z-i)

For (i) both points z_0=+i and z_0=-i lie outside the contour hence integral value is **zero**

For (ii) point z_0=+i lies outside (since $|i + i| = |2i| > 1$) the contour hence its integral value is **zero.** But point z_0=-i lies inside hence Using above formula we get f(z) as $\frac{z^2 - 1}{z - i}$ so integral value comes out to be 2π

So overall $\int_C \frac{z^2 - 1}{z^2 + 1} dz = 0 + 2\pi = \mathbf{2\pi}$

For (iii) points z_0=+i is in whereas point z_0=-i is out. So the integral value comes out to be **-2π.**

For (iv) points z_0=+i is in whereas point z_0=-i is out. So the integral value comes out to be **-2π.**

1. If $f\ z = \sum_{n \geq 0} a_n z^n$ is analytic on and within a circle about the origin, find a_n.

$$f^{\ j}\ z = j! a_j + \sum_{n - j \geq 1} a_n \quad z^{n - j}$$

$$f^{\ j}\ 0 = j! a_j$$

$$\therefore \quad a_n = \frac{f^{\ n}\ 0}{n!} = \frac{1}{2\pi i}\oint \frac{f\ z\ dz}{z^{n+1}}$$

2. In the above case, $|f\ z| \leq M$ on a circle of radius r about the origin, then $|a_n| r^n \leq M$ (Cauchy's inequality)

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Proof:

$$|a_n| = \frac{1}{2\pi}\left|\oint_{|z|=r} \frac{f\ z\ dz}{z^{n+1}}\right| \le M\ \ r\ \ \frac{2\pi r}{2\pi r^{n+1}} \le \frac{M}{r^n}$$

where $M\ \ r\ \ = Max_{|z|=r}\left|f\ \ r\ \right|$.

3. Liouville's theorem: If f(z) is analytic and bounded in the complex plane, it is a constant.
Proof: For any z_0, construct a circle of radius R around z_0,

$$\left|f'\ \ z_0\ \right| = \left|\frac{1}{2\pi i}\oint_R \frac{f\ \ z\ \ dz}{z - z_0\ ^2}\right| \le \frac{M}{2\pi}\frac{2\pi R}{R^2}$$

$$= \frac{M}{R}$$

Since R is arbitrary, let $\mathrm{R} \to \infty$, we have

$$f'\ \ z\ \ = 0, i.e,\ f(z) = const.$$

Conversely, the slightest deviation of an analytic function from a constant value implies that there must be at least one singularity somewhere in the infinite complex plane. Apart from the trivial constant functions, then, singularities are a fact of life, and we must learn to live with them, and to use them further.

7.5 Taylor's Expansion

Suppose we are trying to expand f(z) about $z=z_0$, i.e., $f\ \ z\ \ = \sum_{n-0}^{\infty} a_n\ \ z - z_0\ ^n$, and we have $z=z_1$ as the nearest point for which f(z) is not analytic. We construct a circle C centered at $z=z_0$ with radius $\left|z' - z_0\right| < \left|z_1 - z_0\right|$ (Fig.5)

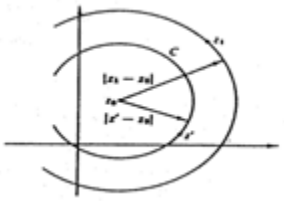


Fig. 5 Fig. 5

From the Cauchy integral formula,

$$f\ \ z\ \ = \frac{1}{2\pi i}\oint_C \frac{f\ \ z'\ \ dz'}{z' - z} = \frac{1}{2\pi i}\oint_C \frac{f\ \ z'\ \ dz'}{z' - z_0\ \ -\ \ z - z_0}$$

$$= \frac{1}{2\pi i}\oint_C \frac{f\ \ z'\ \ dz'}{z' - z_0\ \left[1 -\ \ z - z_0\ /\ \ z' - z_0\ \right]}$$

Here z′ is a point on C and z is any point interior to C. For |t| <1, we note the identity

$$\frac{1}{1-t} = 1 + t + t^2 + \cdots = \sum_{n=0}^{\infty} t^n$$

So we may write

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$$f(z) = \frac{1}{2\pi i}\oint_C \sum_{n=0}^{\infty} \frac{(z-z_0)^n f(z')\,dz'}{(z'-z_0)^{n+1}}$$

$$= \frac{1}{2\pi i}\sum_{n=0}^{\infty} (z-z_0)^n \oint_C \frac{f(z')\,dz'}{(z'-z_0)^{n+1}}$$

$$= \sum_{n=0}^{\infty} (z-z_0)^n \frac{f^n(z_0)}{n!}$$

which is our desired Taylor expansion, just as for real variable power series, this expansion is unique for a given z_0.

Example 1: Expand log (1+z) as Taylor's series expansion about z_0=0

Here f(z)= log(1+z) so we get its derivatives

$$f'(z) = \frac{1}{1+z};\ f''(z) = \frac{-1}{(1+z)^2};\ f'''(z) = \frac{(-1)*(-2)}{(1+z)^3}$$

$$So\ f^n(z) = \frac{(-1)^{n-1}(n-1)!}{(1+z)^n}$$

Evaluate the values of f(z) and its derivatives at z=0 so the series obtained looks like

$$f(z) = z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \ldots\ldots\ldots\ldots + (-1)^{n-1}\frac{z^n}{n!} + \ldots$$

Expand $\frac{1}{z(z^2-3z+2)}$ for the regions

i) 0<$|Z|$ <1 ii) 1<$|Z|$ <2 iii) $|Z|$ >2

Firstly to rewrite the denominator into product of linear terms and then resolving into the partial functions we get

$$f(z) = \frac{1}{2z} - \frac{1}{z-1} + \frac{1}{2(z-2)}$$

i) For **0<$|Z|$ <1 we have**

$f(z) = \frac{1}{2z} + \frac{1}{1-z} - \frac{1}{4(1-z/2)}$ This modification was required to get in the form of basic series of $(1+x)^{-1}$ which can be easily expanded. Also depending on contour region modification in the third term done by taking 2 as common so that z/2 <1 for values of z between 0&1. On expanding each of the above term we get the series as

$$f(z) = \frac{1}{2z} + \frac{3}{4} + \frac{7z}{8} + \frac{15z^2}{16} + \ldots\ldots\ldots$$

ii) For **1<$|Z|$ <2 we have**

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$f(z)=\frac{1}{2z}+\frac{1}{z}(\frac{1}{1-1/z})-\frac{1}{4(1-z/2)}$ This modification done to have terms for which series expansion is known and also depending on the region around which to expand so that for expansion of $(1+x)^{-1}$ we have x<1 which would lead to decrease in the error when compared to actual functional value. We ge

$$f(z)=\left(-\frac{1}{2z}-\frac{1}{z^2}-\frac{1}{z^3}-....\right)-\frac{1}{4}\left(1+\frac{z}{2}+\frac{z^2}{2^2}+\frac{z^3}{2^3}+.....\right)$$

iii) For $|Z|$ **>2 we have**

$f(z)=\frac{1}{2z}+\frac{1}{z}(\frac{1}{1-1/z})+\frac{1}{2z(1-2/z)}$**.** Similar to above reasons modification done and hence can be written in series expansion. We get $f(z)=\frac{1}{2z}-\frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+....\right)+\frac{1}{2z}\left(1+\frac{2}{z}+\frac{2^2}{z^2}+\frac{2^3}{z^3}+.....\right)$

7.6 Analytic continuation

We consider a Taylor series of an analytic function of defined in some restricted region S_1 (Fig.1.10). Then f is analytic inside the circle of convergence C_1, whose radius is given by the distance r_1 from the center C_1 to the nearest singularity of f at z_1. If we choose a point inside C_1, that is farther than r_1 from z_1 and make a Taylor expansion of f about it (z_2 in Fig.1.10), then the circle of convergence C_1 will usually extend beyond C_1. In the overlap region of both circles, f is uniquely defined. In the region of C_2, f(z) is uniquely defined by the Taylor series about the center of C_2 and analytic there, although the Taylor series about the center of C_1 is no longer convergent there. This process is called analytic continuation. It defines the analytic function f in terms of its original definition in C_1 and all its continuations.

For example

$$f\ z\ =1/\ 1+z$$

which has a simple pole at z = -1 and is analytic elsewhere. For |z| < 1, the geometric series expansion

$$\frac{1}{1+z}=1-z+z^2+\cdots=\sum_{n=0}^{\infty}-z^{\ n}$$

Suppose we expand it about z = i, so that

$$f\ z\ =\frac{1}{1+i+\ z-i}=\frac{1}{1+i\left[1+\ z-i\ /\ 1+i\ \right]}$$

$$=\frac{1}{1+i}\left[1-\frac{z-i}{1+i}+\frac{z-i^{\ 2}}{1+i^{\ 2}}+\cdots\right]$$

converges for $|z-i|<|1+i|=\sqrt{2}$ (Fig.1.10)

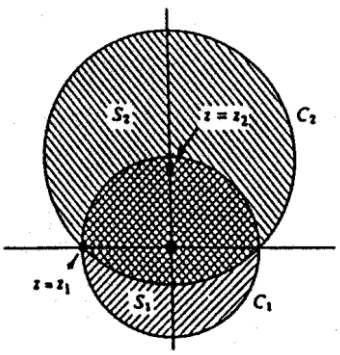


Fig. 1.10

The above three equations are different representations of the same function. Each representation has its own domain of convergence.

7.7 Maclaurin series

For Taylor's series we have infinite terms in the series so the remainder tends to zero. The Taylor's series expansion about $z_0=0$ gives us Maclaurin series i.e.

$$f(z)=\ f(0)+z*\ f\ '(0)+\frac{z^2}{2!}*\ f\ ''(0)+.....$$

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Permanence of Algebraic Form

All our elementary functions, e^z, sin z, and so on can be extended into the complex plane. For example, they can be defined by power-series expansions such as

$$e^z = 1 + z + \frac{z^2}{2!} + \cdots = \sum_{n=0}^{\infty} \frac{z^n}{n!}$$

for the exponential. Such definitions literally constitute an analytic continuation of the corresponding real functions into the complex plane. This result is often referred as permanence of the algebraic form.

7.8 Laurent Series

We frequently encounter functions that are analytic in annular region (Fig.1.11).

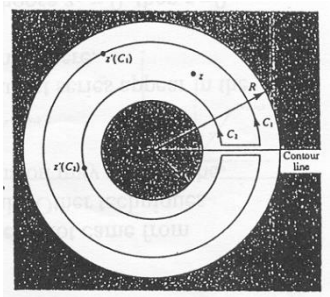


Fig. 1.11

Drawing an imaginary contour line to convert our region into a simply connected region, we apply Cauchy's integral formula for C_2 and C_1, with radii r_2 and r_1, and obtain

$$f(z) = \frac{1}{2\pi i}\left[\oint_{C_1} - \oint_{C_2}\right] \frac{f(z')\,dz'}{z' - z}$$

We let $r_2 \to r$ and $r_1 \to R$, so for C_1, $|z' - z_0| > |z - z_0|$ while for C_2, $|z' - z_0| < |z - z_0|$. We expand two denominators as we did before

$$f(z) = \frac{1}{2\pi i}\left\{\oint_{C_1} \frac{f(z')\,dz'}{z' - z_0\left[1 - (z - z_0)/(z' - z_0)\right]} + \oint_{C_2} \frac{f(z')\,dz'}{z - z_0\left[1 - (z' - z_0)/(z - z_0)\right]}\right\}$$

$$= \frac{1}{2\pi i}\sum_{n=0}^{\infty} (z - z_0)^n \oint_{C_1} \frac{f(z')\,dz'}{(z' - z_0)^{n+1}} + \frac{1}{2\pi i}\sum_{n=0}^{\infty} \frac{1}{(z - z_0)^{n+1}} \oint_{C_2} (z' - z_0)^n f(z')\,dz' \qquad \therefore$$

$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \qquad \text{(Laurent Series)}$$

where

$$a_n = \frac{1}{2\pi i}\oint_C \frac{f(z')\,dz'}{(z' - z_0)^{n+1}}$$

Here C may be any contour with the annular region $r < |z - z_0| < R$ encircling z_0 once in a counterclockwise sense.

Laurent Series need not came from evaluation of contour integrals. Other techniques such as ordinary series expansion may provide the coefficients.

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Let $f\ z\ =\left[z\ z-1\ \right]^{-1}$. If we choose $z_0 = 0$, then r = 0 and R = 1, f(z) diverging at z = 1

$$a_n = \frac{1}{2\pi i}\oint \frac{1}{z'^{n+1}}\frac{dz'}{z'\ z'-1} = \frac{-1}{2\pi i}\oint \sum_{m=0}^{\infty} z'^{\ m}\frac{dz'}{z'^{n+2}}$$

If we employ the polar form,

$$a_n = -\frac{1}{2\pi i}\sum_{m=0}^{\infty}\oint \frac{rie^{i\theta}d\theta}{r^{n+2-m}e^{i\ n+2-m\ \theta}}$$

$$= -\frac{1}{2\pi i}\cdot 2\pi i\sum_{m=0}^{\infty}\delta_{n+2-m,1}$$

$$\therefore \qquad a_n = \begin{cases} -1 & \text{for n} \geq \text{-1} \\ 0 & \text{for n} < \text{-1} \end{cases}$$

The Laurent expansion becomes

$$\frac{1}{z\ z-1} = -\frac{1}{z} - 1 - z - z^2 - z^3 - \cdots = -\sum_{n=-1}^{\infty} z^n$$

7.9 Singularities, Residues and Evaluation of Real Integral

Part I: Singularities

Def. A point $z = a$ is called a **singularity** of a function $f(z)$ if f is not differentiable at $z = a$.

***Examples* 1**

(a) $f(z) = e^{3/z}$; $a = 0$

(b) $f(z) = (z-1)^{-2}$; $a = 1$

(c) $f(z) = \tan z$; $a = {}^{(2n+1)\pi}\!/_{2}$

Classification of the Singularity *a* of *f*

depends on the Laurent Series of f about a, i.e.,

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n.$$

ii) a is called a **Removable singularity** if $c_n = 0$ for all $n < 0$

***Example* 2:** $f(z) = (\sin z)/z$; $a = 0$.

ii) a is called a **Essential singularity** if

$c_n = 0$ for infinitely many $n < 0$

***Example* 3 :** $f(z) = e^{3/z}$; $a = 0$.

iii) a is called a **Simple Pole** if

$c_{-1} \neq 0$ and $c_n = 0$ for all $n < -1$

Career Avenues

***Example* 4:** $f(z) = (z-1)^{-1}$; $a = 1$.

iv) a is called a **Pole of Order *m*** if

$c_{-m} \neq 0$ and $c_n = 0$ for all $n < -m$.

***Example* 5:** $f(z) = (z-1)^{-5}$; $a = 1$ is a pole of order 5 for the function *f*.

Zero of Order *m*

A point $z = a$ is called a **zero of order *m*** of a function $f(z)$ if

$$f^{(j)}(a) = 0,\ j = 0,1,\ldots,m-1 \text{ but } f^{(m)}(a) \neq 0.$$

***Example* 6:** $f(z) = (z-1)^5$; $a = 1$ is a Zero of order 5 for the function *f*.

Note: If $z = a$ is a **zero of order *m*** of a function $f(z)$, then $z = a$ is a pole **of order *m*** for $1/f(z)$.

Test for the Order of a Pole of Rational Functions

$F(z) = f(z) / g(z)$

$z = a$ is a pole **of order *m*** for F if

i. $f(a) \neq 0$.

ii. $z = a$ is a **zero of order *m*** of $f(z)$.

***Exercise*:** Apply this test to the above examples.

Part II: Residue of $f(z)$ at a Singularity $z = a$

is defined with the help of the Laurent Series of f about a, i.e.,

if we have

$$f(z) = \sum_{n=-\infty}^{\infty} c_n (z-a)^n ,$$

then the number c_{-1} is called the **Residue** of $f(z)$ at $z = a$.

Notation: "Residue of $f(z)$ at $z = a$" = **Res(f , a).**

Examples

In Examples 1

(a) $\text{Res}(e^{3/z}, 0) = 3$

(b) $\text{Res}((z-1)^{-2}, 1) = 0$

Note: $\text{Res}((z-1)^{-1}, 1) = 1$

Career Avenues

Method for Finding Res(f, a)

[without Laurent Series]

Case 1: If a is a Simple Pole of f, then

$\text{Res}(f, a) = \lim_{z \to a}(z-a) f(z)$.

***Example*:** $\text{Res}((z-3)(z-1)^{-1}, 1) = -2$

Case 2: If a is a Pole of order m of f, then

$$\text{Res}(f, a) = \frac{1}{(m-1)!}\left(\lim_{z \to a} \frac{d^{m-1}}{dz^{m-1}}(z-a)^m f(z)\right).$$

***Examples*:**

i. $\text{Res}((z-1)^{-2}, 1) = 0$ (Check)

ii. $\text{Res}((z-3)^{-1}(z-1)^{-2}, 1) = -\frac{1}{4}$ (Check)

iii. $\text{Res}((z-3)^{-1}(z-1)^{-2}, 3) = \frac{1}{4}$ (Check)

Note: We can not use this method if a is not a Pole: For example check for $\text{Res}(e^{3/z}, 0)$.

Part II: Residue Theorem

(To Evaluate $\oint_C f(z)dz$ where C is a closed path)

Conditions:

i. C is a simple path in a simply connected domain D.

ii. $f(z)$ is differentiable on and within C except at a finite number of singularities, say $a_1, a_2, \ldots, a_n$ within C.

Conclusion:

$$\oint_C f(z)dz = 2\pi i \sum_{i=1}^{n} \text{Res}(f, a_i)$$

Examples

1. $\int_{|z|=\frac{1}{2}} \frac{dz}{(z-1)(z+2)^2} = 0$ (No singularities within $|z|=1/2$)

2. $\int_{|z|=\frac{3}{2}} \frac{dz}{(z-1)(z+2)^2} = 2\pi i(\text{Res}(f, 1) = \frac{2\pi i}{9}$

3. $\int_{|z|=1} \frac{dz}{z^2+6iz} = 2\pi i(\text{Res}(f, 0) = 2\pi i * \frac{1}{6i} = \frac{\pi}{3}$ ($Singular\ point\ at\ z = -6i\ lie\ outside$)

Career Avenues

Part III: Evaluation of Real Integrals by using Residue Theorem

A. *Type* 1: Real Trigonometric Integral $I=\int_0^{2\pi} F(\cos\theta,\sin\theta)d\theta$

Method(Change of variable): *C*: $z=e^{i\theta}, 0\le\theta\le 2\pi.$

Then: $d\theta=\frac{dz}{iz}, \cos\theta=\frac{z+z^{-1}}{2}, \sin\theta=\frac{z-z^{-1}}{2i}$

(Note: $e^{i\theta}=\cos\theta+i\sin\theta$)

Now use ***Residue Theorem*** to solve

$$I=\oint_C F\left(\frac{1}{2}(z+z^{-1}),\frac{1}{2}(z-z^{-1})\right)\frac{dz}{z}.$$

Example 1: Evaluate $\int_0^{2\pi}\frac{d\theta}{2-\sin\theta}$

Solution: i. Use: (a) $z=e^{i\theta}, 0\le\theta\le 2\pi.$

(b) $d\theta=\frac{dz}{iz}, \sin\theta=\frac{z-z^{-1}}{2i}$

ii. $I=\oint_C \frac{-2dz}{z^2-4iz-1}=\oint_C \frac{-2dz}{(z-a_1)(z-a_1)}\equiv\oint_C f(z)dz$

Here: $a_1=(2-\sqrt{3})i; a_2=(2+\sqrt{3})i$ (Only a_1 is inside *C*)

iii. $\oint_C f(z)dz=2\pi i\,\mathrm{Res}(f,a_1)=\frac{2\pi}{\sqrt{3}}$ (Answer)

$$\left(\mathrm{Res}(f,a_1)=\lim_{z\to a_1}\left[(z-a_1)\frac{-2}{(z-a_1)(z-a_2)}\right]=\frac{1}{i\sqrt{3}}\right)$$

Career Avenues

SET: Practice Questions-2

1. Express the following functions of a complex variable in the form $w = u(x, y) + jv(x, y)$:
   (i) $f(z) = jz + 1$
   (ii) $f(z) = z^2 + 2z + 1$

2. Show that the function $f(z) = z^3$ is analytic

3. Prove that function u(x,y) is Harmonic and hence finds its conjugate harmonic
   (i) y^3-$3x^2y$
   (ii) ln (x^2+y^2)
   (iii) $(x-1)^3$-$3xy^2$+$3y^2$
   (iv) x^3-$3x^2y$-$3x^2$-$3y^2$+1
   (v) e^xcosy

4. Expand the following functions in Taylor series about z=0given point
   (i) ez
   (ii) sin z
   (iii) cos z
   (iv) tan-1 z
   (v) 1/(i+z)
   (vi) ln (1-z)
   (vii) (1+z)-1
   (viii) sec z
   (ix) z/(ez-1)
   (x) sin-1(z)

5. Express the following functions of a complex variable in the form $w = u(x, y) + jv(x, y)$:
   (i) $f(z) = 1/(z-2)$
   (ii) $f(z) = z\bar{z}$

6. Find residue at all singular points for the following functions
   (i) $\dfrac{4-3z}{z^2-z}$
   (ii) $\sin z / z^4$
   (iii) $\dfrac{3z+6}{(z+1)(z^2+16)}$
   (iv) $\dfrac{z+1}{(z-1)^2(z^2+1)}$

7. Evaluate the integral using the residue theorem
   (i) $\int_{|z|=1} \dfrac{dz}{z^2+6iz}$
   (ii) $\int_{|z-i|=1} \dfrac{(z^2-1)dz}{z^2+1}$
   (iii) $\int_{|z|=2} \dfrac{(z^2+z+2)dz}{z^2+1}$

Career Avenues

Solutions:Practice Set-2

1. (i) We have

$$\begin{aligned} f(z) &= jz+1 \\ &= j(x+jy)+1 \\ &= (1-y)+j(x) \end{aligned}$$

(ii) We have

$$\begin{aligned} f(z) &= z^2+2z+1 \\ &= (x+jy)^2+2(x+jy)+1 \\ &= (x^2-y^2+2jxy)+2(x+jy)+1 \\ &= (x^2-y^2+2x+1)+j(2xy+2y) \end{aligned}$$

2. We have

$$\begin{aligned} f(z) &= z^3 \\ &= (x^3-3xy^2)+j(3x^2y-y^3) \end{aligned}$$

So that

$$\frac{\partial u}{\partial x} = 3x^2-3y^2 = \frac{\partial v}{\partial y}$$

$$\frac{\partial v}{\partial x} = 6xy = -\frac{\partial u}{\partial y}$$

The Cauchy-Riemann equations are satisfied so the function is analytic

3. The corresponding conjugate analytic function v(x,y) are

(i) $-3xy^2+x^3+c$

(ii) $2\tan^{-1}(y/x)+c$

(iii) $3x^2y-6yx+3y-y^3+c$

(iv) $3x^2y-y^3+6xy+c$

(v) $e^x\text{siny}+c$

Career Avenues

4.

$(i)\ \ 1+z+z^2/2!+z^3/3!+.....$

$(ii)\ \ z-z^3/3!+z^5/5!-...$

$(iii)\ 1-z^2/2!+z^4/4!-.....$

$(iv)\ \ z-z^3/3+z^5/5-...$

$(v)\ \ -i+z+iz^2-z^3.....$

$(vi)\ \ z+z^2/2+z^3/3+.....$

$(vii)\ 1-z+z^2/2!-z^3/3!+.....$

$(viii) 1-\frac{(-1)}{2!}z^2+\frac{5}{4!}z^4-\frac{6!}{6!}z^6+.....$

$(ix)\ \ 1+\frac{(-1)}{2}z+\frac{1}{6*2!}z^2+\frac{0}{3!}z^3-.....$

$(x)\ \ z+\frac{1}{2}(\frac{z^3}{3})+(\frac{1}{2}*\frac{3}{4})\frac{z^5}{5}+.....$

5. **(i)** $f(z)=\frac{x-2}{(x-2)^2+y^2}+j\frac{-y}{(x-2)^2+y^2}$

(ii) $f(z)=x^2+y^2$

6. (i) Res (f, 0)= -4; Res(f,1)=1

(ii) Res(f,0)=-1/3!

(i) Res (f,-1)=3/17; Res (f,4i)= $\frac{-6+27i}{68}$; Res(f,-4i)= $\frac{-6-27i}{68}$

(ii) Res(f, 1)=-1/2; Res(f, i)=(1+i)/4; Res(f, -i)=(1-i)/4

7. (i) $\pi/3$

(ii) -2π

(iii) 0

Career Avenues

SET: 20

Analytic function

1. The real part of principal value of 4^{4-i} is
   a) $256\,Cos(\ln 4)$
   b) $64\,Cos(\ln 4)$
   c) $16\,Cos(\ln 4)$
   d) $4\,Cos(\ln 4)$

2. Let $i=\sqrt{-1}$ then one value of i^i is
   a) $\sqrt{i}$
   b) -1
   c) $\pi/2$
   d) $e^{-\pi/2}$

3. The principal value of $\log_e i^{\frac{1}{4}}$ is
   a) $i\pi$
   b) $i\pi/2$
   c) $i\pi/4$
   d) $i\pi/8$

4. The function $f(z)=\dfrac{3z}{z^2+4}$ is continuous every where except
   a) $z=\pm 2i$
   b) $z=\pm i$
   c) $z=\pm 3i$
   d) $z=\pm 4i$

5. The locus of the point z saisfying the conditions $\arg\dfrac{z-1}{z+1}=\dfrac{\pi}{3}$ is
   a) A straight line
   b) Circle
   c) A Parabola
   d) None of these

6. Consider the circle $|z-5-5i|=2$ in the complex plane (x,y) with $z=x+iy$. The minimum distance from the origin to the circle is
   a) $5\sqrt{2}-2$
   b) $\sqrt{54}$
   c) $\sqrt{34}$
   d) $5\sqrt{2}$

Career Avenues

7. Consider the function $f(z)=x^2+iy^2$ *and* $g(z)=x^2+y^2+ixy$ *, at* $z=0$

a) f *is analytic but not* g

b) g *is analytic but not* f

c) *Both* f *and* g *are analytic*

d) *Niether* f *nor* g *is analytic*

8. The complex analytic function $f(z)$, *with the imaginary part* $e^x(y\cos y+x\sin y)$ *is*

a) ze^{z+c}

b) $(z+c)e^z$

c) ze^z

d) $(z^2+z)e^{z^2+z}$

9. Which of the following is not the real part of an analytic function ?

a) x^2-y^2

b) $\dfrac{1}{1+x^2+y^2}$

c) x^2-y^2

d) $\cos x\cosh y$

e) $x+\dfrac{x}{x^2+y^2}$.

10. The harmonic conjugate of $u(x,y)=x^2-y^2+xy$ *is*

a) x^2-y^2-xy

b) x^2+y^2-xy

c) $2xy+\dfrac{1}{2}(y^2-x^2)$

d) $\dfrac{1}{2}xy+2\,(y^2-x^2)$

11. The conjugate function of $u=2x(1-y)$ *is*

a) x^2+y^2-2y+C

b) x^2-y^2+2y+C

c) x^2-y^2-2y+C

d) *None of the above*

12. If $f(z)=x^2+iy^2$, then $\dot{f}(z)$ *exist* at all points on the line

a) $x=y$

b) $x=-y$

c) $x=2+y$

d) $y=x+2$

Career Avenues

13. If $u = \sinh x \cos y$ then the analytic function $f(z) = u + iv$ is

a) $\cosh^{-1} z + iC$
b) $\cos hz + iC$
c) $\sin hz + iC$
d) $\sinh^{-1} z + iC$

14. If $v = 2xy,$ then the analytic function $f(z) = u + iz$ is

a) $z^2 + C$
b) $z^{-2} + C$
c) $z^3 + C$
d) $z^{-3} + C$

15. If $u = \dfrac{x-y}{x^2+y^2}$, then analytic function $f(z) = u + iv$ is

a) $z + C$
b) $z^{-1} + C$
c) $(1-i)\dfrac{1}{z} + C$
d) $(1+i)\dfrac{1}{z} + C$

16. For the function $f(z) = \dfrac{z - \sin z}{z^3}$, the point $z = 0$ is

a) A pole of order 3
b) A pole of order 2
c) An essential singularity
d) A removable singularity

17. For the function $\dfrac{\sin z}{z^3}$ of a complex variable z , the point $z = 0$ is

a) A pole of order 3
b) A pole of order 2
c) A pole of order 1
d) Not a singularity

18 If $f(z) = z^3$ then it

a) Has an essential singularity at $z = \infty$
b) Has a pole of order 3 at $z = \infty$
c) Has a pole of order 3 at $z = 0$
d) Is analytic at $z = \infty$

Career Avenues

Solutions

SET: Analytic Functions

1. Ans: a)

Let $x + iy = 4^{4-i}$

Then we get $\log, (x + iy) = (4-i) \log,4$

Or $x + iy = e^{(4-i)\log,4}$

Or $x + iy = e^{4 \log,4}. e^{-i\log, 4}$

Or $x + iy = 4^4 [\cos (\log,4)-i \sin (\log,4)]$

Or $x = 256 \cos (\log,4)$

Hence A is Correct option

2. Ans: d)

We know that

$i^i = e^{i \log i}$

$= e^{i\left(i\frac{x}{2}\right)}$

$= e^{-x/2}$

Hence D is correct option

3. Ans: d)

We have $z = \ln\ i^{1/4} = \frac{1}{4}\log i$

But $i = e^{ix/2}$

Thus $z = \frac{1}{4}\log^{ix/2} = \frac{1}{4} \times \frac{i\pi}{2}$

$= \frac{i\pi}{8}$

Hence D is correct option

4. Ans: a)

We have $f\ x = \frac{3z}{z^2+4} = \frac{3z}{z+2i\quad z-2i}$

Here the denominator is zero when (z +2i) (z-2i) =0

Or $z = \pm 2i$

Thus the function f (z) is continuous except at $z = \pm 2i$

Hence A is correct option

5. Ans: b)

We have $\arg\frac{z-1}{z+1} = \frac{\pi}{3}$

Putting z = x + iy, $\arg\frac{z+iy-1}{x+iy+1} = \frac{\pi}{3}$

Or $\tan^{-1}\frac{y}{z-1} - \tan^{-1}\frac{y}{x+1} = \frac{\pi}{3}, \qquad since \arg\frac{z_1}{z_2} = \arg z_1 - \arg z_2$

Or $\tan^{-1}\frac{\frac{y}{x-1} - \frac{y}{x+1}}{1+\frac{y^2}{x^2-1}} = \frac{\pi}{3}$

Or $\frac{2y}{x^2+y^2-1} = \tan\frac{\pi}{3} = \sqrt{3}$

Or $x^2 + y^2 - 2\sqrt{3}\ y - 1 = 0.$ which is circle

Hence B is correct option

Career Avenues

6. Solu: - a)
Given complex equation is, $|z-5-5i|=2$ which is equation of circle having center at (5,5) and radius 2. The distance of its center form origin is $5\sqrt{2}$. Thus minimum distance from origin is $5\sqrt{2}-2$
Hence A is correct option

7. Ans: c)
We Have $f(x)=x^2+iy^2$ and $g(z)=x^2+y^2+ixy$
Let $f(z)=u+iv$ then $u=x^2$ and $v=y^2$
And also $\frac{\partial u}{\partial x}=2x, \frac{\partial v}{\partial y}=2y, \frac{\partial u}{\partial y}=0$ and $\frac{\partial v}{\partial x}=0$
At z = 0, x = y = 0
Now $\frac{\partial u}{\partial x}=0=\frac{\partial v}{\partial y}$ and $\frac{\partial u}{\partial y}=0=-\frac{\partial v}{\partial x}$
Hence f (x) is analytic
Let $g(z)=a+ib$ then $a=x^2+y^2$ and $b=zy$
At z = 0 x = y = 0
Then $\frac{\partial a}{\partial x}=2x=0, \frac{\partial b}{\partial y}=x=0, \quad \frac{\partial a}{\partial y}=2y=0 \quad$ and $\frac{\partial b}{\partial x}=y=0$
Now $\frac{\partial a}{\partial x}=\frac{\partial b}{\partial y}, \frac{\partial a}{\partial y}=-\frac{\partial a}{\partial x}$
Thus g (z) is also analytic
Hence C is correct option

8. Ans: c)
Let f(x) = u + iv then v = e^x (y cos y + x sin y)
Or $\frac{\partial v}{\partial x}=e^x(y\cos y+x\sin y+\sin y)$
$\frac{\partial u}{\partial y}=e^x(y\cos y+x\sin y+\sin y)$ since f (x) is analytic
Now $\frac{\partial v}{\partial y}=e^x(\cos y-y\sin y+x\cos y)=\frac{\partial u}{\partial x}$
Let $\phi_1(x,y)=\frac{\partial u}{\partial x}$ and $\phi_2(x,y)=\frac{\partial u}{\partial y}$
Then $f(z)=\int \phi_1(z,0)\,dz-i\int \phi_2(z,0)\,dz+C$
$=\int e^z(1+z)\,dz+i\int e^z(0+0+0)+C$
$=\int e^z\,dz+\int ze^z\,dx+C$
$=e^z+ze^z-e^z+C$
$=ze^z$
Hence C is Correct option

9. Ans: b)
Let $\frac{1}{1+x^2+y^2}$ be real part then we have
$u=\frac{1}{1+x^2+y^2}$

Career Avenues

Or $$\frac{\partial u}{\partial x}=\frac{-2x}{1+x^2+y^2\ ^2}=\frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial y}=\frac{-2y}{1+x^2+y^2\ ^2}=-\frac{\partial v}{\partial x}$$

$$dv=\frac{\partial v}{\partial x}dx+\frac{\partial v}{\partial y}dy=\frac{2ydx-2xdy}{1+x^2+y^2\ ^2}$$

From above equation v can't be determined hence u is not a real part of an analytic function
Hence B is correct option

10. Solu: - c)

We have $u=x^2-y^2+xy$

Now $\frac{\partial u}{\partial x}=2x+y=\frac{\partial v}{\partial y}$

and $\frac{\partial u}{\partial y}=-2y+x=-\frac{\partial v}{\partial x}$

$$v=\int\frac{\partial v}{\partial x}dx+\int\frac{\partial v}{\partial y}dy$$

$$=\int\ 2y-x\ dx+\int\ 2x+y\ dy$$

$$=2xy+\frac{1}{2}\ y^2-x^2$$

Hence C is correct option

11. Ans: b)

We have $\frac{\partial u}{\partial x}=2\ 1-y\ ,and\ \frac{\partial^2 u}{\partial x^2}=0$ .. 1

$\frac{\partial u}{\partial y}=-2x\ and\ \frac{\partial^2 u}{\partial y^2}\ 0$ .. 2

Now $\frac{\partial^2 u}{\partial x^2}+\frac{\partial^2 u}{\partial y^2}=0,$ Thus u is harmonic

Let b be the conjugate of u then by Cauchy- Riemann equation

$$dv=-\frac{\partial u}{\partial y}dx+\frac{\partial u}{\partial x}dy$$

$$dv=2xds+2\ 1-y\ dy$$

On integrating, $v=x^2-y^2+2y+C$

Hence B is correct option

12. Ans: a)

Let f(z) = u + iv then $u = x^2$ and $v = y^2$

Now we have $\frac{\partial u}{\partial u}=2x,\frac{\partial u}{\partial y}=0,\frac{\partial u}{\partial y}=0\quad and\ \frac{\partial v}{\partial y}=2y$

We know that $f'\ z\ =\frac{\partial u}{\partial x}-i\frac{\partial u}{\partial y}$ .. 1

and $f'\ z\ =\frac{\partial v}{\partial y}+i\frac{\partial v}{\partial x}$ .. 2

Form equation (1) we get $f'\ z\ =2x$ .. 3

And from equation (2) we get $f'\ z\ =2y$ .. 4

Career Avenues

Now, for existence of f'(z) at any point is necessary that the value of f'(z) most be unique
At that point, whatever be the path of reaching at that point
Form equation (3) and (4) 2x = 2y
Thus f'(z) exists for all points lie on the line x = y
Hence A is correct option.

13. Ans: c)

We have u = sin h x cos y

Then $\frac{\partial u}{\partial x} = \cos h\, x \cos y = \phi\ x, y$

And $\frac{\partial u}{\partial y} = -\sin h\, x \sin y = \phi\ x, y$

By Milne's Method $f'\ z\ = \phi\ z,0\ - \phi\ z,0\ = \cos h\, z - i.0 = \cos hz$

On integrating $f\ x\ = \sin h\, z + constant$

Or $f\ x\ = w = \sin h\, z + iC$

As u does not contain any constant, the constant c is in the function x and thus i.e. in u
Hence C is correct option

14. Ans: a)

We Have $\frac{\partial v}{\partial x} = 2y = h\ x, y\ \ and\ \frac{\partial v}{\partial y} = 2x = g\ \ x, y$

By Milne's Method $f'\ z\ = g\ \ z,0\ + ih\ \ z,0\ = 2z + i\,0 = 2z$

On integrating $f\ \ z\ = z^2 + C$

Hence A is Correct option

15. Ans: d)

We Have $v = \frac{x - y}{x^2 + y^2}$

Now $\frac{\partial v}{\partial y} = \frac{-\ x^2 + y^2\ -\ x - y\ \ 2y}{x^2 + y^2\ \ ^2} = \frac{y^2 - x^2 - 2xy}{x^2 + y^2\ \ ^2} = g\ \ x, y$

$\frac{\partial v}{\partial x} = \frac{x^2 + y^2\ -\ x - y\ \ 2x}{x^2 + y^2} = \frac{y^2 - x^2\ + 2xy}{x^2 + y^2} = h\ \ x, y$

By Milne's Method $f'\ z\ = g\ \ z,0\ + ih\ \ z,0$

$= -\frac{1}{z^2} + i\left(-\frac{1}{z^2}\right) = -\ 1 + i\ \ \frac{1}{z^2}$

On integrating $f\ \ z\ = 1 + i\ \int \frac{1}{z^2} dz + C = 1 + i\ \ \frac{1}{z} + C$

Hence D is correct option

16. Ans: d)

We Have $f\ \ z\ = \frac{z - \sin z}{z^3}$

$= \frac{z - \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - ...\right)}{z^3}$

$= \frac{1}{3!} - \frac{z^2}{5!} + ...$

Career Avenues

Now $\lim_{x\to 0} f\ z = \frac{1}{3!} = \frac{1}{6}$

Hence, z = 0 is a removable singularity
Hence D is correct option

17. Ans: a)

We Have $f\ z = \frac{\sin z}{z^3}$

It can be easily seen that f (z) has a pole of order 3.
Hence A is correct option

18. Ans: b)
It can be easily seen that f (z) – z^3 has a pole of order 3 ar z = ∞
Hence B is correct option

Career Avenues

SET: 21

Cauchy's Theorem

1. If $\oint_c \frac{e^{2z}}{(z+1)^4} dz = ?$ where C is the circle of $|z| = 3$

   a) $\frac{4}{9}\pi i e^{-3}$ b) $\frac{4}{9}\pi i e^{3}$ c) $\frac{4}{3}\pi i e^{-1}$ d) $\frac{8}{3}\pi i e^{-2}$

2. The value of the integral $\oint_c \frac{1-2z}{z(z-1)(z-2)}$ is……. Where $C\, is\, a\, circle\ |z| = 1.5$

   a) $2 + i6\pi$ b) $4 + i3\pi$ c) $1 + i\pi$ d) $i3\pi$

3. $\boldsymbol{f(z_0) = \oint_c \frac{3z^2+7z+1}{z-z_0} dz}$, where $\boldsymbol{C\ is\ a\ circle\ x^2 + y^2 = 4}$

   The value of $f(3) is$

   a) 6
   b) 4i
   c) -4i
   d) 0

4. $\boldsymbol{f(z_0) = \oint_c \frac{3z^2+7z+1}{z-z_0} dz}$, **where** $\boldsymbol{C\ is\ a\ circle\ x^2 + y^2 = 4}$

   The value of $\dot{f}(1-i)$ is

   a) $7(\pi + i2)$
   b) $6(2 + i\pi)$
   c) $2\pi(5 + i13)$
   d) 0

5. The value of $\oint_c \frac{2\cos \pi z}{z^2 - 1} dz$ around a rectangle with vertices at $2 \pm i, -2 \pm i\ is$

   a) 6 b) $i2e$ c) 8 d) 0

6. The value of Integral $\oint_c \frac{sin\pi z^2}{(z-2)(z-1)} dz\ ?Where C is the circle |z| = 3$

   a) $i6\pi$ b) $i2\pi$ c) $i4\pi$ d) 0

7. The value of $\oint_c \frac{cos\pi z}{z-1} = ?Where\ C\ is the circle |z| = 3$

   a) $2i\pi$
   b) $-2i\pi$
   c) $6i\pi^2$
   d) $-6i\pi^2$

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8. value of the $\oint_c \left(z - z^2\right) dz = ?,\ where\, C\, is\, the\, upper\, half\, of\, the\, circle\, |z| = 1$

a) $-2/3$

b) $2/3$

c) $3/2$

d) $-3/2$

9. The value of $\oint_c z^2 e^{\frac{1}{z}}\, dz = ?\, Where\, C\, is\, |z| = 1$

a) $i3\pi$

b) $-i3\pi$

c) $\dfrac{i\pi}{3}$

d) $None\, of\, the\, above$

10. The value of $\oint_c \dfrac{z\, Cos\, z}{\left(z - \dfrac{\pi}{2}\right)^2} dz = ?\, Where\, C\, is\, |z-1| = 1$

a) 6π b) -6π c) $2\pi i$ d) $None\, of\, the\, above$

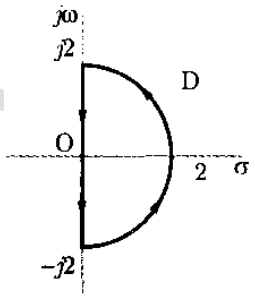
11. The value of $\int_C \dfrac{e^{2z}}{(z-1)^4} dz = ?$

a) $\dfrac{8}{3}\pi i e^2$ b) $\dfrac{8}{3}\pi i$ c) e^{-2} d) $\dfrac{8}{3}\pi i e^{-2}$

12. The value of $\oint_{|z-j|=2} \dfrac{1}{z^2+4} dz$ is

a) $\dfrac{\pi j}{2}$ b) $\dfrac{-\pi}{2}$ c) $\dfrac{-\pi j}{2}$ d) $\dfrac{\pi}{2}$

13. For the given contour D (semi-circle of radius 2) as shown in figure, the value of $\oint_D \dfrac{1}{(s^2 - 1)} ds$ *is*



a) πj b) $-\pi j$ c) π d) $-\pi$

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14. For any circle C enclosing the origin and oriented in counter clock-wise, then the value of $\oint_C \frac{\cos z}{z^2} dz$ *is*

a) $2\pi i$ b) 0 c) $-2\pi i$ d) Undefined

15. The value of $\oint_C \frac{\cot(\pi z)}{(z-1)^2} dz$, where C is a contour $4x^2 + y^2 = 2(Counter\,clock-wise)$

a) 0 b) $-2\pi i$ c) $2\pi i\left(\frac{\pi}{sinh^2\pi} - \frac{1}{\pi}\right)$ d) $\left(\frac{-2\pi i}{sinh^2\pi}\right)$

16. The value of $f(z) = \frac{3z+4}{(z+1)(z+2)}$ along the path $|z| = 3$ is

a) $2\pi i$ b) $4\pi i$ c) $6\pi i$ d) $8\pi i$

17. The value of $\oint_C \frac{z^2}{2z+3} dz\, is \ldots\ldots\ldots..,$ where C is a boundary of sqaure lies along $z = \pm 1 and\ y = \pm 1.$

a) $\frac{\pi i}{4}$ b) $2\pi i$ c) 0 d) $-2\pi i$

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Solutions:

1. Ans: d)

We Have by the derivative of an analytic function that

$$f^{n}(z_0) = \frac{n!}{2\pi i}\oint_c \frac{f(z)\,dz}{(z-z_0)^{n+1}}$$

Or

$$\oint_c \frac{f(z)\,dz}{(z-z_0)^{n+1}} = \frac{2\pi i}{n!} f^{n}(z_0)$$

Taking n = 3

$$\oint_c \frac{f(z)\,dz}{(z-z_0)} = \frac{\pi i}{3} f'''(z_0)$$

We Have

$$\oint_c \frac{e^{2z}dz}{(z+1)^4} = \oint_c \frac{e^{2z}dz}{[z-(-1)]^4}$$

Taking f (x) = e^{3x}, and z_0 = -1 in (1), we have

$$\oint_c \frac{e^{2z}dz}{(z+1)^4} = \frac{\pi i}{3} f'''(-1)$$

Now, f (z) – e^{2z} gives f ‘‘‘ (z) = $8e^{2z}$, then we have. f ‘‘‘ (-1) = $8e^{-2}$

Using this in equation (2) $\oint_c \frac{e^{2z}dz}{(z+1)^4} = \frac{8\pi i}{2}e^{-2}$

Since C is the circle $|z| = 3$ and f (z) is analytic within and on $|z| = 3$,

$$\oint_{|z|=3} \frac{e^{2z}}{(z+1)^4} = \frac{8\pi i}{3}e^{-2}$$

Hence D is correct option

2. Ans: d)

We Have,

$$\frac{1-2z}{z(z-1)(z-2)} = \frac{1}{2z} + \frac{1}{z-1} - \frac{3}{2(z-2)}$$

$$\oint_c \frac{1-2z}{z(z-1)(z-2)}dz = \frac{1}{2}I_1 + I_2 - \frac{3}{2}I_3 \qquad ..(1)$$

Since, z = 0 is the only singularity for $I_1 = \oint_c \frac{1}{z}dz$ and it lies insides $|z| = 15$, Therefore by

Cauchy’s integral formula

$$I_1 = \oint_c \frac{1}{z}dz = 2\pi i \qquad ..(2)$$

Similarly, for $I_2 = \int_x \frac{1}{z-1}dz$, the singular point z =1 lies inside $|z| = 1.5$, therefore

$$I_2 = 2\pi i \qquad ..(3)$$

For $I_3 = \int_c \frac{1}{z-2}dz$, The singular point z = 2 lies outside the circle $|z| = 15$, Thus by

Cauchy’s integral theorem

$$I_3 = \oint_c \frac{1}{z-2}dz = 0 \qquad ..(4)$$

Using equations (2), (3), (4) in (1), we get

$$\oint_c \frac{1-2z}{z(z-1)(z-2)}dz = \frac{1}{2}(2\pi i) + 2\pi i - \frac{3}{2}(0)$$

$$= 3\pi i$$

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Hence D is current option

3. Ans: d)

We have $$f\ 3 = \oint_c \frac{3z^2+7z+1}{z-3}dz$$

Since $z_0 = 3$ is the only singular point of f(3) and it lies outside the circle $x^2 + y^2 = 4$ i.e., $|z| = 2$, Therefore function is analytic everywhere within C.

Thus By Cauchy's theorem $$f\ 3 = \oint_c \frac{3z^2+7z+1}{z-3} = 0$$

Hence D is current option

4. Ans: c)

The point (1-i) lies within circle $|z| = 2$, the distance of 1-I i.e., (1,2) form the origin is $\sqrt{2}$ which is less than 2, the radius of the circle.

Let $\phi\ z\ -3z^2+7z+1$ then by Cauchy's integral formula

$$\oint_x \frac{3z^2+7z+1}{z-z_0}\,dz = 2\pi i\phi\ z_0$$

Or $$f\ z_0\ = z\pi i\phi\ z_0$$

Or $$f'\ z_0\ = 2\pi i\phi\ z_0$$

Or $$\phi\ z\ = 3z^2+7z+1$$

Or $$\phi'\ z\ = 6z+7$$

Thus $$f'\ 1-i\ = 2\pi i\left[6\ 1-i\ +7\right]$$

$$= 2\pi\ 5+13i$$

Hence C is correct option

5. Ans: d)

Let $$I = \oint_c \frac{2}{z^2-1}\cos\pi z dz$$

$$= \oint_c \left(\frac{1}{z-1} - \frac{1}{z+1}\right)\cos\pi z\,dz$$

$$I = \oint_c \left(\frac{\cos\pi z}{z-1} - \frac{\cos\pi z}{z+1}\right)dz$$

$$= 2\pi i\ \cos\pi z\ _{x=-1} - 2\pi i\ \cos\pi z\ _{x=1}$$

Hence D is correct option

6. Ans: d)

We have $$\oint_c \frac{\sin\pi z^2}{z-2\ \ z-1}dz = \oint_c \frac{\sin\pi z^2}{z-2}dz - \oint_c \frac{\sin\pi z^2}{z-1}dz$$

$$= 2\pi i f\ 2\ - 2\pi i f\ 1 \qquad \text{where } f\ z\ = \sin\pi z^2$$

Thus $$f\ 2\ = \sin 4\pi = 0$$

And $$f\ 1\ = \sin\pi = 0$$

Hence D is correct option

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7. Ans: b)

Let f(z) = cos π z than f(z) analytic within and on |z| = 3, Noe by Cauchy's integral formula

$$f\ z_0 = \frac{1}{2\pi i}\oint_c \frac{f\ z}{z - z_0} dz$$

$$\oint_c \frac{f\ z}{z - z_0} dz = 2\pi i f\ z_0$$

Taking f(z) = cos π z, z_0 = 1, we have

$$\oint_{|x|=3} \frac{\cos \pi z}{z - 1} = 2\pi i f\ 1$$

$$= 2\pi i \cos \pi = -2\pi i$$

Hence B is correct option

8. Ans: b)

Given contour C is the circle $|z| = 1$ *i.e.*, $z = e^{\theta}$ *or* $dz = ie^{\theta}\, d\theta$, Now for upper half of the circle, $0 \le \theta \le \pi$

$$\oint_c z - z^2\ dz = \int_{\theta=0}^{\pi} e^{\theta} - e^{2\theta}\ ie^{\theta}\, d\theta$$

$$= i\int_0^{\pi} e^{2\theta} - e^{3\theta}\ d\theta$$

$$= i\left[\frac{e^{2\theta}}{2i} - \frac{e^{3\theta}}{3i}\right]_0^{\pi} = \frac{2}{3}$$

Hence B is correct option

9. Ans: c)

We have $f\ x = z^2 e^{1/2} = z^2\left(1 + \frac{1}{z} + \frac{1}{2!z^2} + \frac{1}{3!z^3} + ...\right)$

$$= z^2 + z^2 + \frac{1}{2} + \frac{1}{6z} + ...$$

The only pole of f (z) is at z= 0, which lies within the circle $|z| = 1$

Thus $\oint_c f\ z\ dz = 2\pi i\ \ residue\, at\, z = 0$

Now, residue of f (z) at z = 0 is the coefficient of $\frac{1}{z}$ *i.e.*, $\frac{1}{6}$

$$\oint_c f\ z\ dz = 2\pi i \times \frac{1}{6} = \frac{1}{3}\pi i$$

Hence C is Correct Option

10. Ans: d)

The f (z) has a pole at $z = \frac{\pi}{2}$ of order 2, By Cauchy's residue theorem

$$\oint_C f\ z\ dz = 2\pi i\left(residue\, at\, z = \frac{\pi}{2}\right)$$

Now, Residue at $z = \frac{\pi}{2} is$ $= \lim_{z \to \frac{\pi}{2}} \frac{d}{dz}\left|\left(z - \frac{\pi}{2}\right)^2 f\ z\right|$

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$$= \lim_{z \to \frac{\pi}{2}} \frac{d}{dz} \; z \cos z$$

$$= \lim_{z \to \frac{\pi}{2}} \; \cos z - z \sin z \; = -\frac{\pi}{2}$$

$$\oint_c f \;\; z \;\; dz = 2\pi i \times \left(-\frac{\pi}{2} \right) = -\pi^2 i$$

Hence D is correct option

11. Ans: d)

We Have f(z) – e^{2z} and a = -1

Now $f \text{'''} \;\; z \; = 8e^{2z} \; and \; f \text{'''} \;\; -1 \; = 8e^{-2}$

By Cauchy's integral formula $\oint_c f \;\; z \;\; dz = \frac{2\pi i}{3!} 8e^{-2}$

$$= \frac{2\pi i}{6} \times 8e^{-2}$$

$$= \frac{8}{3} \pi i e^{-2}$$

Hence D is correct option

12. Ans: d)

We Have $$\oint_{|z-j|=2} \frac{1}{z^2+4} dz = \oint_{|z=j|=2} \frac{1}{z+2i \quad z-2i} dz$$

P (0,2) lies inside the circle $|z-j|$=2 and P (0,-2) does not be Thus By Cauchy's integral formula

$$I = 2\pi i \lim_{x \to 2i} \; z - 2i \; \frac{1}{z+2i \quad z-2i}$$

$$= \oint_c \frac{2\pi i}{2i+2i} = \frac{\pi}{2}$$

Hence D is correct Option

13. Ans: a)

We know That $\oint_D \frac{1}{s^2-1} ds = 2\pi j$ [sum of residues]

Singular points are at $s = \pm 1$ but only s = + 1 lies inside the given contour, Thus Residue At s = +1 is

$$\lim_{s \to 1} \; s-1 \;\; f \;\; s \; = \lim_{s \to 1} \; s-1 \; \frac{1}{s^2-1} = \frac{1}{2}$$

$$\oint_D \frac{1}{s^2-2} ds = 2\pi j \left(\frac{1}{2} \right) = \pi j$$

Hence A is Correct Option

14. Ans: b)

Residue at Origin $R = \lim_{z \to 0} \frac{d}{dz} \left(z^2 \frac{\cos z}{z^2} \right)$

Or $R = \lim_{z \to 0} \; -\sin z \; = 0$

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$$I=\oint_c \frac{\cos z}{z^2}dz=2\pi i.R=0$$

Hence B is Correct option

15. Ans: d)

Given integral I has a double pole at z = i

Residue at z – i, is

$$=\frac{1}{2-1\ !}\lim_{z\to i}\frac{d}{dz}\left[\ z-i\ ^2\ \frac{\cot\pi z}{z-i\ ^2}\right]$$

$$=\lim_{z\to i}\frac{d}{dz}\left[\frac{d}{dz}\cot\ \pi z\ \right]$$

$$=-\frac{\pi}{\sin h^2\pi}$$

Now

$$I=2\pi i\times-\frac{\pi}{\sin h^2\pi}=-\frac{2\pi^2 i}{\sin h^2\pi}$$

Hence D is correct option

16. Ans: c)

We Have

$$f\ \ a\ =\frac{3s+4}{s+1\ \ s+2}$$

f(s) has two poles enclosed by contour, therefore

$$I=2\pi i\left[\lim_{z\to-1}\left(\frac{3s+4}{s+2}\right)+\lim_{z\to-2}\left(\frac{3s+4}{s+1}\right)\right]$$

$$=\ =2\pi i\ 1+2\ \ =6\pi i$$

Hence C is correct option

17. Ans: c)

There is a simple pole at z = $-\frac{3}{2}$ but this is not enclosed within the given square

$x=\pm1, y=\pm1 as|z|=\sqrt{2}$

Thus

$$\oint_c \frac{z^2}{2z+3}dz=2\pi i\ \ 0\ \ =0$$

Hence C is correct option

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SET:22

Taylor's and Laurent's Series

1. For the function $f(z) = \dfrac{z-1}{z+1}$, The Taylor series expansion about the point $z = 0$ will be

a) $1+2\left(z+z^2+z^3+\ldots\ldots\ldots\ldots..\right)$

b) $-1-2\left(z-z^2+z^3+\ldots\ldots\ldots\ldots..\right)$

c) $-1+2\left(z-z^2+z^3+\ldots\ldots\ldots\ldots..\right)$

d) *None of the above*

2. For the function $f(z)=\sin z$ *, the Taylor series exapnsion about the point* $z=\dfrac{\pi}{4}$ *is*

a) $\dfrac{1}{\sqrt{2}}\left[1+\left(z-\dfrac{\pi}{4}\right)-\dfrac{1}{2!}\left(z-\dfrac{\pi}{4}\right)^2-\ldots\ldots\ldots\right]$

b) $\dfrac{1}{\sqrt{2}}\left[1+\left(z-\dfrac{\pi}{4}\right)+\dfrac{1}{2!}\left(z-\dfrac{\pi}{4}\right)^2+\ldots\ldots\ldots\right]$

c) $\dfrac{1}{\sqrt{2}}\left[1+\left(z-\dfrac{\pi}{4}\right)-\dfrac{1}{2!}\left(z-\dfrac{\pi}{4}\right)^2-\ldots\ldots\ldots\right]$

d) *None of the above*

3. For the function $f(z)=\dfrac{1}{z+1}$ *, the Taylor series exapnsion about the point* $z=1$ *is*

a) $\dfrac{-1}{2}\left[1-\dfrac{1}{2}(z-1)+\dfrac{1}{2^2}(z-1)^2\ldots\ldots..\right]$

b) $\dfrac{1}{2}\left[1-\dfrac{1}{2}(z-1)+\dfrac{1}{2^2}(z-1)^2\ldots\ldots..\right]$

c) $\dfrac{1}{2}\left[1+\dfrac{1}{2}(z-1)+\dfrac{1}{2^2}(z-1)^2\ldots\ldots..\right]$

d) *None of the above*

4. For the function $f(z)=\dfrac{1}{z\left(e^z-1\right)}$ *, the Laurent's series expansion for* $|z|<2$ *is*

a) $\dfrac{1}{z^2}+\dfrac{1}{2z}+\dfrac{1}{12}+6z+\dfrac{1}{720}z^2+\ldots\ldots\ldots$

b) $\dfrac{1}{z^2}-\dfrac{1}{2z}+\dfrac{1}{12}-\dfrac{1}{720}z^2+\ldots\ldots\ldots$

c) $\dfrac{1}{z}+\dfrac{1}{12}+\dfrac{1}{634}z^2+\dfrac{1}{720}z^3+\ldots\ldots\ldots$

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d) *None of the above.*

5. For the function $f(z)=\dfrac{1}{z(z-1)(z-2)}$, *the Laurent's series expansion for* $|z+1|<1$ *is*

a) $(z-1)^{-1}+(z-1)+(z-1)^2+(z-1)^3+\ldots.$

b) $(z-1)^{-1}-(z-1)+(z-1)^2-(z-1)^3+\ldots.$

c) $-(z-1)^{-1}+(z-1)-(z-1)^2+(z-1)^3-\ldots.$

d) $-(z-1)^{-1}-(z-1)-(z-1)^3-(z-1)^5-\ldots.$

6. For the function $f(z)=\dfrac{1}{(z-1)(z-2)}$, *the Laurent's series expansion for* $1<|z|<2$ *is*

a) $\dfrac{1}{z}+\dfrac{2}{z^2}+\dfrac{3}{z^3}+\ldots\ldots.$

b) $\ldots-z^{-3}-z^{-2}-z^{-1}-\dfrac{1}{2}-\dfrac{1}{4}z-\dfrac{1}{8}z^2-\dfrac{1}{18}z^3-\ldots\ldots.$

c) $\dfrac{1}{z^2}+\dfrac{3}{z^2}+\dfrac{7}{z^4}+\ldots\ldots.$

d) *None of these*

7. For the function $f(z)=\dfrac{1}{3-2z}$ in power of $(z-3)$ and the radius of convergence of the series so obtained are

a) $-\dfrac{1}{3}\left[1-\dfrac{2}{3}(z-3)+\dfrac{4}{9}(z-3)^2-\ldots\right],\ |z-3|<3$

b) $-\dfrac{1}{3}\left[1-\dfrac{2}{3}(z-3)+\dfrac{4}{9}(z-3)^2-\ldots\right],\ |z-3|<\dfrac{3}{2}$

c) $\dfrac{1}{3}\left[1+\dfrac{2}{3}(z-3)+\dfrac{4}{9}(z-3)^2-\ldots\right],\ |z-3|<\dfrac{3}{2}$

d) $\dfrac{1}{3}\left[1-\dfrac{2}{3}(z-3)-\dfrac{4}{9}(z-3)^2-\ldots\right],\ |z-3|<\dfrac{3}{2}$

8. For the function $f(z)=\dfrac{1}{z-1}-\dfrac{1}{z-2}$, the coefficient o

$\dfrac{1}{z^2}$ *in the laurent's series expansion for* $|z|>2$ *is*

a) -1

b) 0

c) 1

d) 2

9. The value of integral $\int_0^{2\pi}\dfrac{d\theta}{13-5\sin\theta}$ is equal to

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a) $-\pi/6$

b) $-\pi/12$

c) $\pi/12$

d) $-\pi/12$

10. The value of $\int_0^{2\pi} e^{\left(r^{i\theta}-i\theta\right)} d\theta \, equals \, to$

a) $2\pi i$

b) 2π

c) π

d) $i\pi$

11. The value of Integral $\int_0^{2\pi} \frac{d\theta}{2+\cos\theta}$ is

a) $-2\pi/\sqrt{2}$

b) $2\pi/3$

c) $2\pi\sqrt{2}$

d) $-2\pi\sqrt{3}$

12. The residue of $z\cos\frac{1}{z} \; at \; z=0 \; is$

a) $-1/3$

b) $-1/2$

c) $1/3$

d) $1/2$

13. The value of residue of the function $f(z)=\frac{1-e^{2z}}{z^4} \; at \, its \; pole \, is$

a) $-2/3$

b) $-4/3$

c) $4/3$

d) $2/3$

14. The residue of $\frac{\sin z}{z^8} \; at \, z=0 \; is$

a) 0

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b) $\frac{1}{7!}$

c) - $\frac{1}{7!}$

d) *None of these*

15. The residue of $f(z) = \cot z$ *at any of its poles is*

a) 0

b) 1

c) $\sqrt{3}$

d) *None of these*

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Solutions:

1. Ans: c)

$f(z)=\frac{z-1}{z+1}=1-\frac{2}{z+1} \quad \Rightarrow f(0)=-1 \text{ and } f(1)=0$

$f'(z)=\frac{2}{(z+1)^2} \quad \Rightarrow f'(0)=2;$

$f''(z)=\frac{-4}{(z-1)^2} \quad \Rightarrow f''(0)=-4;$

$f'''(z)=\frac{12}{(z+1)^4} \quad \Rightarrow f'''(0)=12; \text{ and so on}$

Now, Taylor series is given by

$$f(z)=f(z_0)+(z-z_0)f'(z_0)+\frac{(z-z_0)^2}{2!}f''(z_0)+\frac{(z-z_0)^3}{3!}f'''(x_0)+\ldots$$

Taking $z_0=0$, $f(z)=-1+(z)(2)+\frac{z^2}{2!}(-4)+\frac{z^3}{2!}(12)+\ldots$

$=-1+2z-2x^2+2z^3\ldots.$

$=-1+2(z-z^2+z^3\ldots.$

Hence C correct option

2. Ans: a)

$f(z)=\sin z \quad \Rightarrow f\left(\frac{\pi}{4}\right)=\sin\frac{\pi}{4}=\frac{1}{\sqrt{2}}$

$f'(z)=\cos z \quad \Rightarrow f'\left(\frac{\pi}{4}\right)=\frac{1}{\sqrt{2}}$

$f''(z)=-\sin z \quad \Rightarrow f''\left(\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}}$

$f'''(z)=-\cos z \quad \Rightarrow f'''\left(\frac{\pi}{4}\right)=-\frac{1}{\sqrt{2}} \text{ and so on}$

Taylor series is given by

$$f(z)=f(z_0)+(z-z_0)f'(z_0)+\frac{(z-z_0)^2}{2!}f''(z_0)+\frac{(z-z_0)^3}{3!}f'''(z_0)+\ldots$$

Taking $z_n=\frac{\pi}{4}$

$$f(x)=\frac{1}{\sqrt{2}}+\left(z-\frac{\pi}{4}\right)\frac{1}{\sqrt{2}}+\frac{\left(z-\frac{\pi}{4}\right)^2}{2!}\left(-\frac{1}{\sqrt{2}}\right)+\frac{\left(z-\frac{\pi}{4}\right)^3}{3!}\left(-\frac{1}{\sqrt{2}}\right)+\ldots$$

$$=\frac{1}{\sqrt{2}}\left[1+\left(z-\frac{\pi}{4}\right)-\frac{1}{2!}\left(z-\frac{\pi}{4}\right)^2-\frac{1}{3!}\left(z-\frac{\pi}{4}\right)^3\ldots\right]$$

Hence A is correct option

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3. Ans: b)

$$f(z)=\frac{1}{z+1} \qquad \Rightarrow f(1)=\frac{1}{2}$$

$$f'(z)=\frac{-1}{(z+1)^2} \qquad \Rightarrow f'(1)=\frac{-1}{4}$$

$$f''(z)=\frac{2}{(z+1)^3} \qquad \Rightarrow f''(1)=\frac{1}{4}$$

$$f'''(z)=\frac{-6}{(z+1)^4} \qquad \Rightarrow f'''(1)=-\frac{3}{8} \text{ and so on}$$

Taylor series is

$$f(z)=f(z_0)+(z-z_0)f'(z_0)+\frac{(z-z_0)^2}{2!}f''(z_0)+\frac{(z-z_0)^3}{3!}f'''(z_0)+$$

Taking $z_0=1$

$$f(z)=\frac{1}{2}+(z-1)\left(\frac{-1}{4}\right)+\frac{(z-1)^2}{2!}\left(\frac{1}{4}\right)+\frac{(z-1)^3}{3!}\left(-\frac{3}{8}\right)+..$$

Or $$f(z)=\frac{1}{2}\left[1-\frac{1}{2}(z-1)+\frac{1}{2^2}(z-1)^2-\frac{1}{2^3}(z-1)^3+....\right]$$

Hence B is correct option

4. Ans: b)

$$f(z)=\frac{1}{z^2}-\frac{1}{2z}+\frac{1}{12}+0.z-\frac{1}{720}z^2+..$$

Hence B is correct option

5. Ans: d)

For $|z-1|<1$ *let* $z-1=u$, *that gives* $z=u+1$ *and* $|u|<1$

We have $$f(z)=\frac{1}{z(z-1)(z-2)}$$

$$=\frac{1}{2z}-\frac{1}{z-1}+\frac{1}{2}\frac{1}{z-2}$$

$$=\frac{1}{2}\frac{1}{u+1}-\frac{1}{u}+\frac{1}{2}\frac{1}{u-1} \qquad z-1=u$$

$$=\frac{1}{2}(1+u)^{-1}-u^{-1}-\frac{1}{2}(1-u)^{-1}$$

$$=\frac{1}{2}(1-u+u^2-u^3+\ldots)-u^{-1}-\frac{1}{2}(1+u+u^2+u^3+\ldots)$$

$$=\frac{1}{2}(-2u-2u^3-\ldots)-u^{-1}=-u-u^3-u^5-\ldots-u^{-1}$$

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6. Ans: b)

We have $f(z)=\frac{1}{(z-1)(z-2)}=\frac{1}{z-2}-\frac{1}{z-1}$

Since, $|z|>1 \text{ or } \frac{1}{|z|}<1 \text{ and } |z|<2 \text{ or } \frac{|z|}{2}<1 \text{ so we get}$

$$\frac{1}{z-1}=\frac{1}{z\left(1-\frac{1}{z}\right)}=\frac{1}{z}\left(1-\frac{1}{z}\right)^{-1}=\frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\ldots\right)$$

and $$\frac{1}{z-2}=\frac{-1}{2}\left(1-\frac{z}{2}\right)^{-1}=-\frac{1}{2}\left[1+\frac{z}{2}+\frac{z^2}{4}+\frac{z^3}{9}+\ldots\right]$$

From equation 1 *we get*

$$f(z)=-\frac{1}{2}\left(1+\frac{z}{2}+\frac{z^2}{2}+\frac{z^3}{9}+..\right)-\frac{1}{z}\left(1+\frac{1}{z}+\frac{1}{z^2}+\frac{1}{z^3}+\ldots\right)$$

Or $$f(z)=\ldots-z^{-3}-z^{-2}-z^{-1}-\frac{1}{2}-\frac{1}{4}z-\frac{1}{8}z^2-\frac{1}{18}z^3-\ldots$$

Hence B is correct option

7. Ans: b)

We have

$$f(x)=\frac{1}{3-2z}$$
$$=\frac{1}{-3\left[1+\frac{2}{3}(z-3)\right]}$$
$$=-\frac{1}{3}\left[1+\frac{2}{3}(z-3)\right]^{-1}$$
$$=-\frac{1}{3}\left[1-\frac{2}{3}(z-3)+\frac{4}{9}(z-3)^2-..\right]$$

and radius of concergence is $|z-3|<\frac{3}{2}$

Hence B is correct option

8. Ans: a)

We have

$$f(z)=\frac{1}{z-1}-\frac{1}{z-2},|z|>2$$
$$=\frac{1}{z}\left[\frac{1}{\left(1-\frac{1}{z}\right)}-\frac{1}{\left(1-\frac{2}{z}\right)}\right]$$
$$=\frac{1}{z}\left[\left(1-\frac{1}{z}\right)^{-1}-\left(1-\frac{2}{z}\right)^{-1}\right]$$
$$=\frac{1}{z}\left[\left(1+\frac{1}{z}+\frac{1}{z^2}+..\right)-\left(1+\frac{2}{z}+\frac{4}{z^2}\right)\right]$$
$$=-\frac{1}{z^2}-\frac{3}{z^2}$$

Thus Coefficent of $\frac{1}{z^2}=1$

Hence A is correct option

Career Avenues

9. Ans: c)

$$I=\int_0^{2\pi}\frac{d\theta}{13-5\sin\theta}=\int_0^{2\pi}\frac{d\theta}{13-5\left(\frac{e^{\theta}-e^{-\theta}}{2i}\right)}$$

Let $e^{\theta}=z$ *or* $d\theta=\frac{dz}{iz}$ *and C will be unit circle* $|z|=1$

We get

$$I=\int_0^{2\pi}\frac{2d\theta}{26+5ie^{\theta}-5ie^{-\theta}}$$

$$=\oint_c\frac{2}{26+5iz-\frac{5i}{z}}\frac{dz}{iz}$$

$$=2\oint_c\frac{dz}{5-5z^2+26iz}$$

Poles are given by $-5z^2+28iz+5=0$,

Or

$$z=\frac{-26i\pm\sqrt{-676+100}}{-10}$$

$$=\frac{-26i\pm24i}{-10}=\frac{i}{5},5i$$

Only $z=\frac{i}{5}$ *lies inside C*

Thus

$$I=2\pi i\ \text{Residue at } z=i/5$$

$$=2\pi i\lim_{z\to\frac{i}{5}}\left[\left(z-\frac{i}{5}\right)\times\frac{1}{5z-i\quad -z+5i}\right]$$

$$=2\pi i\times\frac{1}{5\left(-\frac{i}{5}+5i\right)}=\frac{\pi}{12}$$

Hence C is correct Option

10. Ans: b)

Let

$$I=\int_0^{2\pi}e^{\,e^{\theta-i\theta}}\,d\theta=\int_0^{2\pi}e^{e\theta}e^{-i\theta}d\theta$$

Let $z=e^{i\theta}$ *or* $d\theta=\frac{dz}{iz}$ *and C will be* $|z|=1$, *we get*

$$I=\oint_c e^{-1}z^{-1}\frac{dz}{iz}=\frac{1}{i}\oint_c\frac{e^x}{z^2}dz$$

The Only Pole in C is z = 0 of order 2.

Thus

$$I=2\pi i\left(\frac{1}{i}\right)\ \text{residue at } z=0$$

$$=2\pi\left[\frac{1}{1!}\frac{d^1}{dz^1}z^2\frac{e^z}{z^2}\right]_{z=0}$$

$$=2\pi\left[\frac{e^2}{1}\right]_{z=0}=2\pi$$

Hence B is correct option

Career Avenues

11. Ans: b)

Let $z = e^{\theta}$ *then we have* $d\theta = \dfrac{-idz}{z}$ *where* $z \le \theta \le 2\pi$

and $\cos\theta = \dfrac{1}{2}\left(z + \dfrac{1}{z}\right)$

$$\int_0^{2\pi} \frac{d\theta}{2+\cos\theta} = \oint_c \frac{\frac{-idz}{z}}{2+\frac{1}{2}\left(z+\frac{1}{z}\right)} \quad \text{where } C \text{ is } |z| = 1$$

$$= -2i \oint_c \frac{dz}{z^2+4z+1}$$

Let $f(z) = \dfrac{1}{z^2+4z+1}$ then f(z) has poles at z = -2 + $\sqrt{3}, -2-\sqrt{3},$ out of these only $z = -2+\sqrt{3}$ lies inside the circle $|z| = 1$

Thus $\oint_c f(z)\,dz = 2\pi i$ *Residue at* $z = -2+\sqrt{3}$

Residue at $z = -2+\sqrt{3}$ *is* $= \lim_{z \to -2+\sqrt{3}} \left(z+2-\sqrt{3}\right) f(z)$

$$= \lim_{z \to -2+\sqrt{3}} \frac{1}{z+2+\sqrt{3}} = \frac{1}{2\sqrt{3}}$$

$$\oint_c f(z)\,dz = 2\pi i \times \frac{1}{2\sqrt{3}} = \frac{\pi i}{\sqrt{3}}$$

$$\int_c^{2\pi} \frac{d\theta}{2+\cos\theta} = -2i \times \frac{\pi i}{\sqrt{3}} = \frac{2\pi}{\sqrt{3}}$$

Hence B is correct option

12. Ans: b)

We have $f(z) = z\cos\dfrac{1}{z}$

$$= z\left(1 - \frac{1}{2!}\frac{1}{z^2} + \frac{1}{4!}\frac{1}{z^4} - \ldots\right)$$

$$= z - \frac{1}{2z} + \frac{1}{24x^3} - \ldots$$

Residue of f (z) at z= 0 is the coefficient of $\dfrac{1}{z}$ *i.e*, $-\dfrac{1}{2}$.

Hence B is correct option

Career Avenues

13. Ans: b)

The function $f(z)=\frac{1-e^{2x}}{z^4}$ *has a pole at* $z=0$ *of order* 4.

Residue of $f(z)$ *at* $z=0$ *is*

$$=\frac{1}{n-1!}\lim_{z\to z_0}\frac{d^{n-1}}{dz^{n-1}}(z-z_0)^n f(z)$$

$$=\frac{1}{4-1!}\lim_{x\to 0}\frac{d^3}{dz^3}\left[z^4.\left(\frac{1-e^{2i}}{z^4}\right)\right]$$

$$=\frac{1}{3!}\lim_{z\to 0}\frac{d^3}{dz^3}(1-e^{2z})=\frac{-1}{3!}\lim_{z\to 0}8e^{2z}$$

$$=\frac{=8}{6}(e^0)=\frac{-8}{6}=\frac{-4}{3}$$

Hence B is correct option

14. Ans: c)

The function $\frac{\sin z}{z^8}$ has a pole of order 8 at z = 0

$$=\frac{1}{8-1!}\left[\frac{d^{8-1}}{dz^{8-1}}(z-0)^8\frac{\sin z}{z^8}\right]$$

Thus residue is

$$=\frac{1}{7!}\left[\frac{d^7}{dz^7}(\sin z)\right]_{z=0}$$

$$=\frac{1}{7!}(-\cos z)_{z=0}=-\frac{1}{7!}$$

Hence C is correct option

15. Ans: b)

We have $f(z)=\cot z=\frac{1}{\tan z}$

For poles, $\tan z=0$ $z=n\pi$ *where* $n=0,\pm1,\pm2,...$

Residue is $f(z)$ *is* $=\lim_{z\to n\pi}(z-n\pi)\frac{\cos z}{\sin z}$

$$=\lim_{z\to n\pi}\left[\frac{\cos z}{(-1)^n}\times\frac{z-n\pi}{\sin(z-n\pi)}\right]$$

$$=\frac{(-1)^n}{(-1)^n}=1$$

Hence B is Correct option

Career Avenues

Past GATE Questions

1. For the function of a complex Variable $W = \ln Z \ \left(Where W = u + jv \, an \, Z = x + jy\right)$, the $u = constant \, lines$ get mapped in z-plane as
   a) Set of Radial straight lines
   b) Set of Concentric Circles
   c) Set of Confocal hyperbolas
   d) Set of Confocal ellipses **(G-2006, 2-Marks)**

2. The value of contour integral $\oint_{|z-j|} \frac{1}{z^2+4} dz$ in positive sense is
   a) $j\pi/2$
   b) $-\pi/2$
   c) $-j\pi/2$
   d) $\pi/2$ **(G-2006, 2-Marks)**

3. If the Semi- Contour D of Radius 2 is as shown in the figure, then the value of the Integral $\oint_D \frac{1}{s^2-1} ds$ is
   a) $j\pi$
   b) - $j\pi$
   c) $-\pi$
   d) π **(G-2007, 2-Marks)**

4. The equation $sin(z) = 10$ has
   a) No real or complex solution
   b) Exactly two distinct complex solutions
   c) A unique solution
   d) An infinite number of complex solutions. **(G-2008, 1-Marks)**

5. The residue of the function $f(z) = \frac{1}{(z+2)^2 (z-2)^2}$ $at \, z = 2$ is
   a) $-\frac{1}{32}$
   b) $-\frac{1}{16}$
   c) $\frac{1}{16}$
   d) $\frac{1}{32}$ **(G-2008, 2-Marks)**

Career Avenues

6. If $f(z)=C_0+C_1z^{-1}$, then $\oint_{\substack{Unit\\Circle}} \frac{1+f(z)}{z}dz$ is given by

a) $2\pi C_1$
b) $2\pi(1+C_0)$
c) $2\pi jC_1$
d) $2\pi j(1+C_0)$ **(G-2009, 1-Marks)**

7. The residues of a complex function $X(z)=\frac{1-2z}{z(z-1)(z-2)}$ at its poles are

a) $\frac{1}{2}, -\frac{1}{2}\ and\ 1$
b) $\frac{1}{2}, \frac{1}{2}\ and\ -1$
c) $\frac{1}{2}, 1\ and\ -\frac{3}{2}$
d) $\frac{1}{2}, -1 and\ \frac{3}{2}$ **(G-2010, 2-Marks)**

8. The value of Integral $\oint_C \frac{-3z+4}{(z^2+4z+5)}dz$, Where C is the $Circle|z|=1$ is given by

a) 0
b) $1/10$
c) $4/5$
d) 1 **(G-2011, 1-Marks)**

9. If $x=\sqrt{-1}$, then the value of x^x is

a) $e^{-\pi/2}$
b) $e^{\pi/2}$
c) x
d) 1 **(G-2012, 1-Marks)**

10. Given $f(z)=\frac{1}{z+1}-\frac{1}{z+3}$ if C is a counter clockwise path in z-plane

Such that $|z+1|=0$, then the value of $\frac{1}{2\pi j}\oint_C f(z)dz$ is

a) -2
b) -1
c) 1
d) 2 **(G-2012, 1-Marks)**

Career Avenues

11. The real part of an analytic function $f(z)\,where\,z = x + jy$ is given by $e^{-y}\cos(x)$. The Imaginary part of $f(z)$ is

a) $e^{y}\cos(x)$.
b) $e^{-y}\sin(x)$.
c) $-e^{y}\sin(x)$.
d) $-e^{-y}\sin(x)$. **(G-2014, 2-Marks)**

12. Let $z = x + iy$ be a complex variable. Consider that contour integration is performed along the unit circle in anticlockwise direction. Which one of the following statements is not true?

a) The residue of $\frac{z}{z^2-1}\,at\,z=1\,is\,\frac{1}{2}$

b) $\oint_C z^2\,dz = 0$

c) $\frac{1}{2\pi i}\oint_C \frac{1}{z}dz = 1$

d) $\overline{z\left(Complex\,conjugate\,of\,z\right)is\,analytical\;\;function}$ **(G-2015, 1-Marks)**

13. Let $f(z)=\frac{az+b}{cz+d}$ *If* $f(z_1)=f(z_2)$ *for all* $z_1 \neq z_2$, $a=2.b=4\,and\,c=5$, *then* ***d*** *should be equal to……..* **(G-2015, 1-Marks)**

14. If denotes the counter clockwise unit circle, then value of the contour integral $\frac{1}{2\pi j}\oint_C Re(z)\,dz$ *is….* **(G-2015, 1-Marks)**

15. If C is a circle of radius r with centre z_0 , in the complex $z-plane$ and if n is a non-zero integer, then $\oint_C \frac{dz}{(z-z_0)^{n+1}}$ equals

a) $2\pi nj$
b) 0
c) $\frac{\pi j}{2\pi}$
d) $2\pi n$ **(G-2015, 1-Marks)**

16. In the following integral, the contour C encloses the points $2\pi j$ and $-2\pi j$

$-\frac{1}{2\pi}\oint_C \frac{\sin z}{(z-2\pi j)^3}dz$ the value of the itegral is…… **(G-2016, 2-Marks)**

17. Consider the complex valued function $f(z) = 2z^3 + b|z|^3\;where\;z$ is any complex variable. The value of b for the function $f(z)$ is analytic is …….. **(G-2016, 1-Marks)**

Career Avenues

18. The value of the integral $\frac{1}{2\pi j}\oint_C \frac{e^z}{z-2}dz$ along a closed contour c in anti-clockwise direction for

$the\ point\ z_0 = 2\ inside\ the\ conour\ C, and$

$the\ point\ z_0 = 2\ outside\ the\ conour\ C\ respectively\ are,$

a) $(i)2.72 \quad (ii)0$

b) $(i)7.39 \quad (ii)0$

c) $(i)0 \quad (ii)2.72$

d) $(i)0 \quad (ii)7.39$ **(G-2016, 2-Marks)**

19. For $f(z)=\frac{\sin(z)}{z^2}$, the residue of the pole at $z=0$ is …….. **(G-2016, 1-Marks)**

20. The residue of a function $f(z)=\frac{1}{(z-4)(z+1)^3}$ are

a) $\frac{-1}{27} and \frac{-1}{125}$

b) $\frac{-1}{125} and \frac{-1}{125}$

c) $\frac{-1}{27} and \frac{1}{5}$

d) $\frac{-1}{125} and \frac{-1}{5}$ **(G-2017, 1-Marks)**

21. An integral I over a counetr-clockwise circle C is given by

$$I=\oint_C \frac{z^2-1}{z^2+1}e^z\,dz$$

If C is defined as $|z|=3$, then the value of I is

a) $-\pi i \sin(1)$

b) $-2\pi i \sin(1)$

c) $-3\pi i \sin(1)$

d) $-4\pi i \sin(1)$ **(G-2017, 2-Marks)**

Career Avenues

Answers

1 b)	2 d)	3 a)	4 a)	5 a)	6 d)
7 c)	8 a)	9 a)	10 c)	11 b)	12 d)
13) 10	14) 0	15 b)	16) -133.87	17) 0	18 b)
19) 1	20 b)	21 d)			

Solutions

1. b)

$W = \ln z$

$= \log_e z$

$u + jv = \log_e (x + iy)$

$= \frac{1}{2}\log\left(x^2 + y^2\right) + i\tan^{-1}\left(\frac{y}{x}\right)$

u is constant,

$\therefore \quad \frac{1}{2}\log\left(x^2 + y^2\right) = C$

$x^2 + y^2 = C$ [equation of circle having same centre (0,0)]

2. d)

$$\frac{1}{z^2+4} = \frac{1}{z+2j \quad z-2j}$$

Only pole $0.2j$ *lies inside the circle* $|z-j|=2$

$\therefore$ *By cauchy's integral formula*

$$1 = \oint_{|z-ji=2|} \frac{1}{z^2+4}dz$$

$$= \frac{2\pi j}{2j+2j} = \frac{\pi}{2}$$

3. a)

$$\oint \frac{1}{s^2-1}ds = \oint \frac{1}{s+1 \quad s-1}ds$$

$= 2\pi j \times$ *Sum of residue*

residue at pole s $= -1 is\ 0$

residue at pole s $= 1 is$

$$= \lim_{s\to 1} \frac{s-1}{s-1 \quad s+1}$$

$$= \frac{1}{2}$$

$$\Rightarrow \quad \oint \frac{1}{s^2-1}ds = 2\pi j \times \frac{1}{2} = j\pi$$

Career Avenues

4. a)

$$\frac{e^{iz}-e^{-iz}}{2i}=10$$

$$e^{iz}-\frac{1}{e^{iz}}=20i$$

$$e^{iz}{}^{2}-20ie^{iz}-1=0$$

$$e^{iz}=\frac{20i\pm\sqrt{-396}}{2}=\frac{20i\pm6i\sqrt{11}}{2}$$

$$e^{iz}=10i\pm3\sqrt{11}\,i \qquad iz=\ln\left[i\ 10\pm3\sqrt{11}\ \right]$$

$$iz=\ln 1+i\left(\frac{\pi}{2}\pm2n\pi\right)+\ln\ 10\pm3\sqrt{11} \qquad =\ln i+\ln\ 10\pm3\sqrt{11}$$

$$iz=i\left(\frac{\pi}{2}\pm2n\pi\right)+\ln\ 10\pm3\sqrt{11}$$

Therefore infinite number of complex solutions

5. a)

Since $\lim_{z\to2}\left[\ z-2\ ^{2}\ f\ \ z\ \right]$ *is finite and non − zero*

f(z) has a pole order two at z = 2 the residue at z = a is given for a pole order n as

$$Res\ f\ \ a\ =\frac{1}{n-1\ !}\frac{d^{n-1}}{dz^{n-1}}\left[\ z-a\ ^{n}\ f\ \ z\ \right]_{z=a}$$

$$Here\ n=2\ \ pole\ of\ order\ 2\ \ \&\ a=2$$

$$\therefore Res\ f\ \ 2\ =\frac{1}{1!}\frac{d}{dz}\left[\ z-2\ ^{2}\ f\ \ z\ \right]_{z=2}$$

$$=\frac{d}{dz}\left[\ z-2\ ^{2}\ \frac{1}{z+2\ ^{2}\ z-2\ ^{2}}\right]_{z=2}$$

$$=\frac{d}{dz}\left[\frac{1}{z+2\ ^{2}}\right]_{z=2}$$

$$=\left[-2\ z+2\ ^{-3}\right]_{z=2}$$

$$=\frac{-2}{2+2\ ^{3}}=-\frac{1}{32}$$

6. d)

$$f\ \ z\ =c_0+c_1\ z^{-1}$$

$$\oint\frac{1+f\ \ z}{z}dz=?$$

unit circle

if has one pole at origin

$$so\ \oint_{z\to0}\frac{z\ 1+f\ \ z}{z}dz=2\pi\ j\ \ 1+f\ \ 0$$

$$=2\pi\ j\ \ 1+c_0$$

Career Avenues

7. d)

Given that,

$$X\ z = \frac{1-2z}{z\ z-1\ \ z-2}$$

$$Res\ 0 = \lim_{z\to 0} 1.\left[\frac{1-2z}{z-1\ \ z-2}\right] = \frac{1}{2}$$

$$Res\ 1 = \lim_{z\to 1} 1.\left[\frac{1-2z}{z\ z-2}\right] = \frac{-1}{1\ -1} = 1$$

$$Res\ 2 = \lim_{z\to 2} 1.\left[\frac{1-2z}{z\ z-1}\right]$$

$$= \frac{1-4}{2\ 2-1} = \frac{-3}{2}$$

8. a)

$$I = \oint_c \frac{-3z+4}{z^2+4z+5}dz$$

$$= 2\pi\ \ sum of\ residues$$

$$Poles of\ \frac{-3z+4}{z^2+4z+5}\ are given by$$

$$z^2+4z+5=0$$

$$z = \frac{-4\pm\sqrt{16-20}}{2} = \frac{-4\pm 2i}{2}$$

$$= -2\pm i$$

Since the poles lie outside the circle |z| =1
So f (z) is analytic inside the circle |z| = 1

Hence $\oint_c f\ z\ dz = 2\pi i\ 0 = 0$

9. a)

x = i, then in polar coordinates

$$x = \cos\frac{\pi}{2} + i\sin\frac{\pi}{2} = e^{\frac{\pi}{2}i}$$

$$Now,\quad x^x = j^i = e^{\pi i/2\ \ t} = e^{t^2\pi/2} = e^{-\pi/2}$$

10. c)

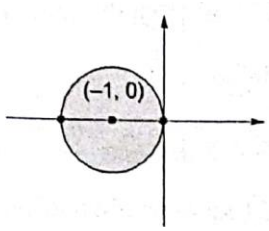
$$f\ z = \frac{1}{z+1} - \frac{2}{z+3}$$

$$= \frac{z+3\ -2\ z+1}{z+1\ \ z+3}$$

$$= \frac{-z+1}{z+1\ \ z+3}$$

Poles are at -1 and -3 i.e, (-1,0) and (-3,0) form figure below of |z+1| = 1,
We see that (-1,0) is inside the circle and (-3,0) is outside the circle

Career Avenues



Residue theorem says $\frac{1}{2\pi j}\oint_c f\ z\ dz$ = Residue of those poles which are inside C. so the required integral $\frac{1}{2\pi j}\oint_c f\ z\ dz$ is given by the residue of function at pole (-1,0) (which is inside the circle)

11. b)

$Z = x + iy$

$f\ z = u + iv$

$u = e^{-y}\cos\ x$

$\frac{\partial u}{\partial x} = -e^{-y}\sin\ x$

For analytical function

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$

$\Rightarrow \frac{\partial v}{\partial y} = -\theta^{-y}\sin\ x$

Integrating w.r.t.y

$V = e^{-y}\sin\ x$

12. d)

$f\ z = \bar{z} = x - iy$

$u = x \quad v = -y$

$u_x = 1 \quad v_x = 0$

$u_y = 0 \quad v_y = -1$

$u_x \neq v_y$ *i.e.C – R not satisfied* $\bar{z}$ *is not analytic function*

13. 10

$f\ z_1 = \frac{az_1 + b}{cz_1 + d}$

$f\ z_2 = \frac{az_2 + b}{cz_2 + d}$

$\frac{az_1 + b}{cz_1 + d} = \frac{az_2 + b}{cz_2 + d}$

$acz_1z_2 + bcz_2 + adz_1 + bd = acz_1z_2 + bcz_1 + adz_2 + bd$

$bc\ z_2 - z_1 = ad\ z_2 - z_1$

$z_2 \neq z_1$

$\Rightarrow \quad bc = ad$

$d = \frac{bc}{a} = \frac{4 \times 5}{2} = 10$

Career Avenues

14. 0

$\text{Re}|z| = \frac{z + \bar{z}}{2}$

Since there is no pole inside unit circle, so Residue at poles is zero

$\Rightarrow \frac{1}{2\pi i}\oint_c \text{Re}|z|\,dz = 0$

15. b)

By Cauchy integral formula

$$\oint \frac{f(z)}{(z - z_0)^{n+1}}\,dz = \frac{2\pi i\, f^*(z_0)}{n!}$$

$$\oint \frac{dz}{(z - z_0)^{n+1}} = \frac{2\pi i}{n!}.0 = 0$$

16. -133.87

$$I = -\frac{1}{2\pi \int_c \frac{\sin z}{(z - 2\pi/)^3}\,dz}$$

$$= -\frac{1}{2\pi} \times \frac{2\pi j\, f''(2\pi l)}{2!}$$

$f(z) = \sin z$

$f'(z) = \cos z$

$f''(z) = -\sin z$

$$I = -\frac{1}{2\pi} \times 2\pi j \frac{-\sin(2\pi j)}{2}$$

$$= -\frac{1}{2}\sin h\, 2\pi = -133.87$$

17. 0

Given that f(z) is analytic

Which is possible only when b = 0

Since $|z^3|$ is differentiable at the origin but not analytic

$2z^3$ is analytic everywhere

$\therefore$ $f(z) = 2z^3 + b|z^3|$ is analytic

Only when b = 0

18. b)

i $z_0 = 2 - lies\, inside\, C,$

so $Res\, f(z) = \lim_{z \to 2}(z - 2).\frac{e^z}{z - 2}$

$= e^2 = 7.38$

$$\frac{1}{2\pi i}\int_c \frac{e^z}{z - 2}\,dz = 2\pi i.\frac{1}{2\pi i}(7.39) = 7.39$$

ii $z_0 = -2\, lies\, out\, side\, C\, then\, Res\, f(z) = 0$

$$so \int_c \frac{e^z}{z - 2}\,dz = 2\pi i \frac{1}{2\pi i}(0) = 0$$

Career Avenues

19. 1

Residue of $\frac{\sin z}{z^2}$

$= \text{coefficient of } \frac{1}{z} in \left\{ \frac{z - \frac{z^3}{3!} + \frac{z^5}{5!} - -}{z^2} \right\}$

$= \text{coefficient of } \frac{1}{z} in \left\{ \frac{1}{z} - \frac{z}{3!} + \frac{z^3}{5!} - - - \right\}$

$=1$

20 b)

Residue at z = 1=-1 is

$$= \lim_{z \to -1} \frac{1}{2!} \frac{d^2}{dz^2} \left(z-1^{\,3} \frac{1}{z-4 \quad z+1^{\,3}} \right)$$

$$= \lim_{z \to -1} \frac{1}{2!} \left(\frac{2}{z-4^{\,3}} \right) = \frac{1}{-1-4^{\,3}} = \frac{-1}{125}$$

Residue at $z = 4$ *is*

$$= \lim_{z \to 4} \; z-4 \; \frac{1}{z-4 \quad z+1^{\,3}} = \frac{1}{4+1^{\,3}} = \frac{1}{125}$$

21. d)

Poles are $z^2 + 1 = 0$

$z = \pm i$

z = i lies inside |z| = 3

z = -i lies inside |z| =3

Residue at z = i is

$$= \lim_{z \to i} \; z-i \; \frac{z^2-1}{z-i \quad z+i} e^z$$

$$= \frac{-1-1}{2i} e^i = ie^i$$

Residue at z = -i is

$$= \lim_{z \to -i} \; z+i \; \frac{z^2-1}{z-i \quad z+i} e^z$$

$$= \frac{-1-1}{-2i} e^{-i} = \frac{1}{i} e^{-i} = -ie^{-i}$$

By residues theorem

$$I = 2\pi i \; ie^i - ie^{-i}$$

$$= -2\pi \; e^i - e^{-i}$$

$$= -2\pi \; \cos 1 + i \sin 1 - \cos 1 + i \sin 1$$

$$= -2\pi \; 2i \sin 1 = -4\pi i \sin \; 1$$

Career Avenues